Agrobusiness

Kate, Anastasiia, Max

2026-04-02

Hypotheses and objectives description

This project investigates the drivers of winter wheat yield volatility across Ukrainian oblasts using panel data for 2015–2024. The core research question is: which meteorological factors actually determine yield outcomes, and which are redundant for practical data collection purposes? The goal is to provide agribusinesses with a parsimonious, empirically grounded framework — identifying what weather data is mandatory to collect and what can be safely dropped.

Research Hypotheses

1. Traditional Significance and Input Factors:

H_0: Meteorological factors have no statistically significant effect on wheat yield.

H_1: Key weather variables — particularly extreme temperature events and precipitation — have significant effects on yield, with the relationship following a non-linear pattern where both heat and cold extremes are detrimental.

2. Seasonal Relevancy (The "Window" Question):

H_0: Weather aggregated over the full calendar year is no less predictive than weather aggregated over the growing season only.

H_1: The growing season window (September–July) captures the agronomically relevant stress periods more precisely than full-year averages, yielding better model fit.

3. Data Granularity and Extreme Day Classification:

H_0: Counting extreme temperature days does not provide more predictive power than simple mean temperature.

H_1: The frequency of extreme events — days above the critical heat threshold and days below the critical cold threshold — is a more accurate predictor of yield shocks than mean temperature alone, justifying the collection of daily rather than weekly data.

4. Redundancy and Variable Selection:

H_0: All weather variables (radiation, ET_0, dewpoint, temperature, precipitation) contribute independent predictive information.

H_1: Solar radiation, ET_0, and dewpoint are strongly collinear with temperature metrics and precipitation, making them redundant. A parsimonious model retaining only temperature stress metrics, precipitation, wind, and soil moisture performs equivalently.

5. Soil Quality as a Moderator:

H_0: Soil properties have no effect on yield once regional fixed effects are controlled for.

H_1: Static soil properties — particularly organic matter and sand content — affect average yield levels and moderate the sensitivity of yield to weather shocks.

Data Description and Exploratory Analysis

1. Introduction

This study investigates the impact of weather factors on winter wheat yield in Ukraine using panel data for 2015–2024 at the oblast level. The dependent variable is yield in centners per hectare. Key explanatory variables are derived from daily temperature and soil moisture data following a theoretically motivated aggregation scheme. Soil properties and area variables serve as controls.

---

2. Data Description

This section describes all data sources used in the study: crop statistics (annual, oblast level), daily weather records, static soil properties, and derived weather metrics constructed following the research hypothesis.

2.1 Crop Data (annual, by oblast)

File	Variable	Units	Explanation
`crop_capacity.csv`	Yield	centners/ha	Dependent variable. Already calculated by statistical authorities.
`harvested_area.csv`	Harvested area	hectares	Area from which crop was actually harvested.
`sowing_area.csv`	Sown area	hectares	Area planted (may differ from harvested due to losses).

```
library(tidyverse)
library(lubridate)
library(patchwork)
```

```r
data_path <- "."
years     <- as.character(2015:2025)

META_COLS <- c("indicator", "region", "unit", "scale",
               "decimals", "special", "note_ts", "note_obs")

read_panel <- function(file, value_name) {
  read_csv(
    file.path(data_path, file),
    col_names      = c(META_COLS, years),
    skip           = 1,
    show_col_types = FALSE
  ) |>
    dplyr::select(region, all_of(years)) |>
    pivot_longer(cols = all_of(years), names_to = "year",
                 values_to = value_name) |>
    mutate(
      year         = as.integer(year),
      !!value_name := suppressWarnings(as.numeric(.data[[value_name]]))
    ) |>
    filter(region != "    ")
}

read_ukraine <- function(file, value_name) {
  read_csv(
    file.path(data_path, file),
    col_names      = c(META_COLS, years),
    skip           = 1,
    show_col_types = FALSE
  ) |>
    filter(region == "    ") |>
    dplyr::select(region, all_of(years)) |>
    pivot_longer(cols = all_of(years), names_to = "year",
                 values_to = value_name) |>
    mutate(year = as.integer(year),
           !!value_name := suppressWarnings(as.numeric(.data[[value_name]])))
}

yield     <- read_panel("crop_capacity.csv", "yield_cpha")
harv      <- read_panel("harvested_area.csv", "harv_kha") |>
               mutate(harv_ha = harv_kha * 1000) |>
               dplyr::select(-harv_kha)
sown      <- read_panel("sowing_area.csv", "sown_kha") |>
               mutate(sown_ha = sown_kha * 1000) |>
               dplyr::select(-sown_kha)

ukr_yield <- read_ukraine("crop_capacity.csv", "yield_cpha")

region_groups <- tribble(
  ~region,              ~group,
  "     ",          "West",
  "       ",       "West",
  "   -      ",  "West",
```

```
  "     ",          "West",
  "     ",          "West",
  "       ",      "West",
  "     ",        "West",
  "     ",        "West",
  "     ",          "Centre",
  "         ",   "Centre",
  "     ",        "Centre",
  "    ",           "Centre",
  "       ",     "Centre",
  "     ",        "Centre",
  "    ",           "Centre",
  "     ",        "Centre",
  "     ",          "Centre",
  "       ",      "Centre",
  "    ",           "East",
  "     ",        "East",
  "     ",          "East",
  "       ",      "East",
  "    ",           "East",
  "     ",        "East"
)

GROUP_COLORS <- c(West = "#2c7bb6", Centre = "#d7191c", East = "#1a9641")

theme_small_multiple <- function(base_size = 9) {
  theme_minimal(base_size = base_size) +
    theme(
      panel.grid.major  = element_line(colour = "grey92", linewidth = 0.3),
      panel.grid.minor  = element_blank(),
      panel.border      = element_blank(),
      panel.background  = element_rect(fill = "white", colour = NA),
      plot.background   = element_rect(fill = "white", colour = NA),
      axis.line.x       = element_line(colour = "grey70", linewidth = 0.4),
      axis.ticks.x      = element_line(colour = "grey70", linewidth = 0.3),
      axis.ticks.y      = element_blank(),
      axis.text.x       = element_text(size = 7, colour = "grey40",
                                       margin = margin(t = 2)),
      axis.text.y       = element_text(size = 7, colour = "grey40"),
      axis.title        = element_blank(),
      strip.text        = element_text(face = "bold", size = 8,
                                       colour = "grey20", margin = margin(b = 3)),
      strip.background  = element_blank(),
      strip.placement   = "outside",
      legend.position   = "none",
      plot.title        = element_text(face = "bold", size = 11,
                                       colour = "grey10", margin = margin(b = 4)),
      plot.subtitle     = element_text(size = 8.5, colour = "grey40",
                                       margin = margin(b = 8)),
      plot.caption      = element_text(size = 7, colour = "grey55",
                                       hjust = 0, margin = margin(t = 6)),
      plot.margin       = margin(10, 12, 8, 10)
    )
```

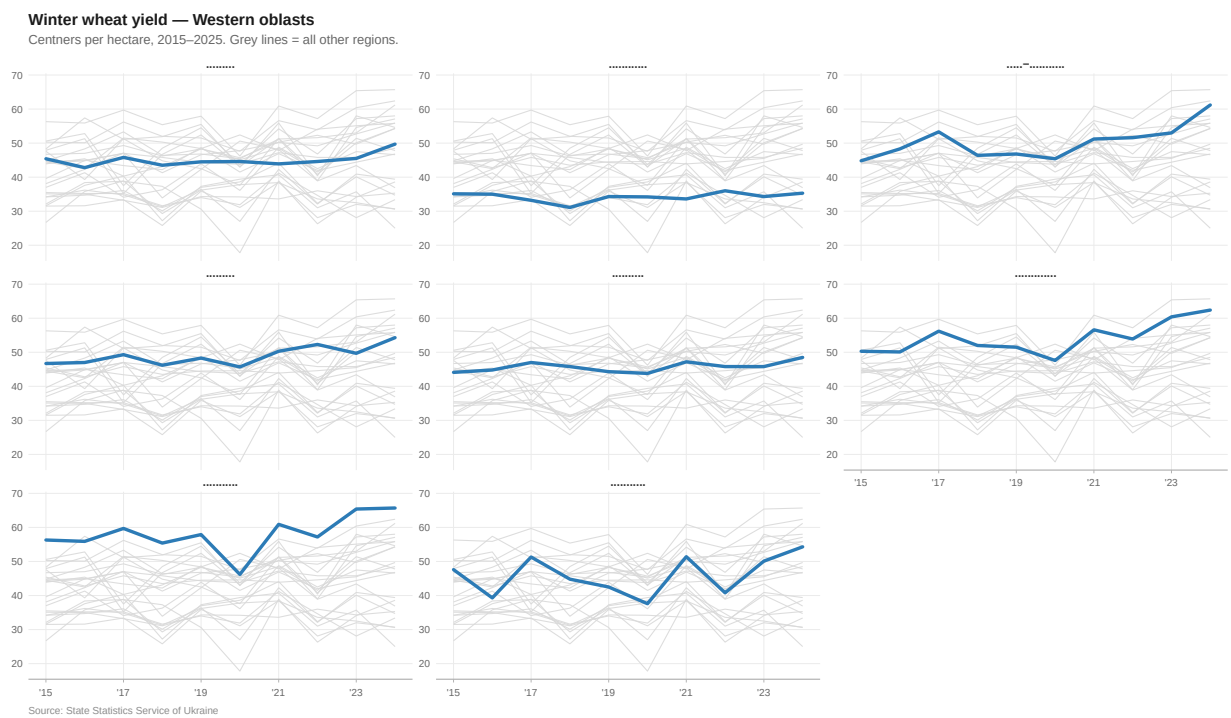
```r
}

plot_small_multiple <- function(df, group_map, value_col,
                                group    = NULL,
                                title    = "",
                                subtitle = "",
                                caption  = "") {

  df <- df |>
    left_join(group_map, by = "region") |>
    rename(value = !!sym(value_col))

  focal_regions <- if (!is.null(group)) {
    group_map |> filter(group == !!group) |> pull(region)
  } else {
    unique(df$region)
  }

  focal_color <- if (!is.null(group)) GROUP_COLORS[[group]] else "grey30"

  bg_data <- tibble(facet = focal_regions) |>
    cross_join(
      df |> filter(!is.na(value)) |> dplyr::select(region, year, value)
    )

  focal_data <- df |>
    filter(region %in% focal_regions, !is.na(value)) |>
    mutate(facet = region)

  ggplot() +
    geom_line(
      data    = bg_data |> filter(region != facet),
      mapping = aes(x = year, y = value, group = region),
      colour  = "grey85", linewidth = 0.35, alpha = 0.9
    ) +
    geom_line(
      data    = focal_data,
      mapping = aes(x = year, y = value, group = facet),
      colour  = focal_color, linewidth = 1.1, lineend = "round"
    ) +
    facet_wrap(~facet, scales = "free_y") +
    scale_x_continuous(
      breaks = c(2015, 2017, 2019, 2021, 2023, 2025),
      labels = c("'15", "'17", "'19", "'21", "'23", "'25")
    ) +
    scale_y_continuous(expand = expansion(mult = c(0.05, 0.10))) +
    labs(title = title, subtitle = subtitle, caption = caption) +
    theme_small_multiple()
}


p_yield_west <- plot_small_multiple(
  yield, region_groups, "yield_cpha", "West",
```

```r
  title    = "Winter wheat yield - Western oblasts",
  subtitle = "Centners per hectare, 2015-2025. Grey lines = all other regions.",
  caption  = "Source: State Statistics Service of Ukraine"
)

p_yield_centre <- plot_small_multiple(
  yield, region_groups, "yield_cpha", "Centre",
  title    = "Winter wheat yield - Central oblasts",
  subtitle = "Centners per hectare, 2015-2025. Grey lines = all other regions.",
  caption  = "Source: State Statistics Service of Ukraine"
)

p_yield_east <- plot_small_multiple(
  yield, region_groups, "yield_cpha", "East",
  title    = "Winter wheat yield - Eastern oblasts",
  subtitle = "Centners per hectare, 2015-2025. Grey lines = all other regions.",
  caption  = "Source: State Statistics Service of Ukraine"
)


loss_rate <- yield |>
  left_join(sown |> dplyr::select(region, year, sown_ha), by = c("region", "year")) |>
  left_join(harv |> dplyr::select(region, year, harv_ha), by = c("region", "year")) |>
  mutate(
    loss_rate = (sown_ha - harv_ha) / sown_ha,
    loss_rate = if_else(loss_rate < 0, 0, loss_rate),
    loss_rate = if_else(loss_rate > 0.5, 0.5, loss_rate)
  ) |>
  filter(!is.na(loss_rate))

p_loss_west <- plot_small_multiple(
  loss_rate, region_groups, "loss_rate", "West",
  title    = "Crop loss rate - Western oblasts",
  subtitle = "(Sown area - Harvested area) / Sown area. 0% = perfect harvest.",
  caption  = "Source: State Statistics Service of Ukraine"
)

p_loss_centre <- plot_small_multiple(
  loss_rate, region_groups, "loss_rate", "Centre",
  title    = "Crop loss rate - Central oblasts",
  subtitle = "(Sown area - Harvested area) / Sown area. 0% = perfect harvest.",
  caption  = "Source: State Statistics Service of Ukraine"
)

p_loss_east <- plot_small_multiple(
  loss_rate, region_groups, "loss_rate", "East",
  title    = "Crop loss rate - Eastern oblasts",
  subtitle = "(Sown area - Harvested area) / Sown area. 0% = perfect harvest.",
  caption  = "Source: State Statistics Service of Ukraine"
)


oblast_area_km2 <- tribble(
```

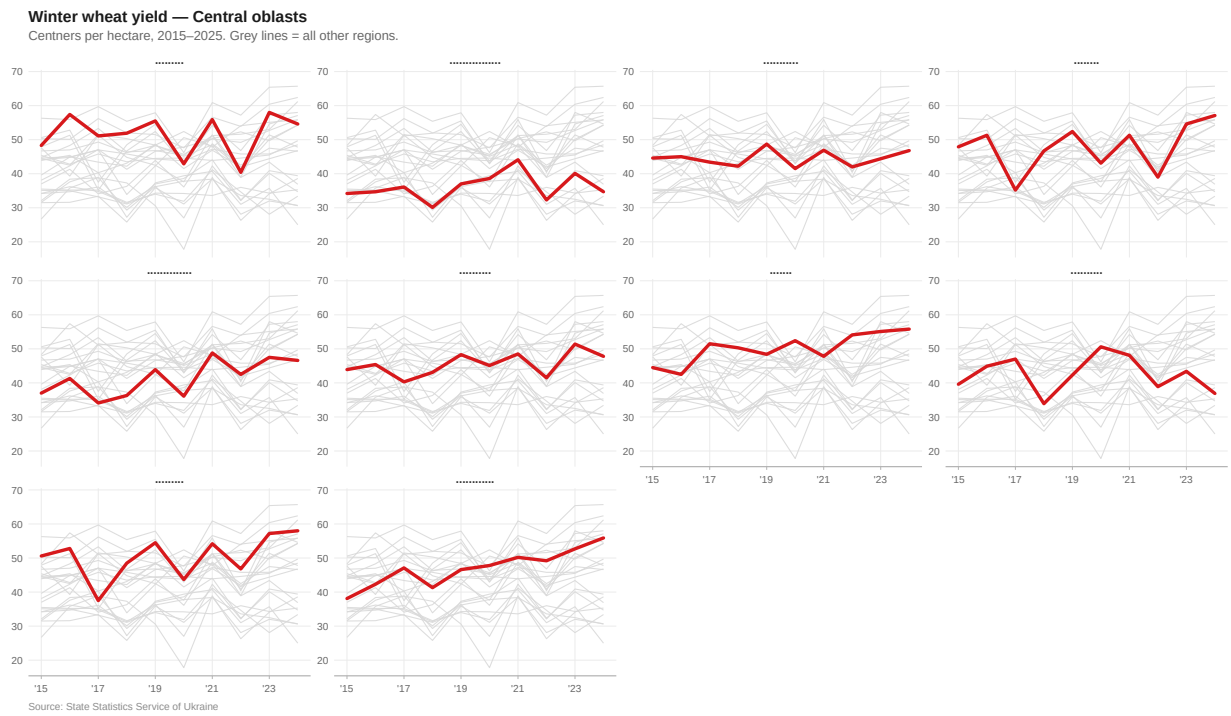
```
  ~region,              ~total_km2,
  "     ",          26513,
  "     ",          20144,
  "         ",   31914,
  "     ",           26517,
  "      ",        29832,
  "       ",       12777,
  "      ",         27180,
  "   -      ",  13900,
  "     ",           28131,
  "        ",     24588,
  "     ",          26683,
  "     ",          21833,
  "       ",       24598,
  "    ",            33310,
  "      ",         28748,
  "      ",         20047,
  "    ",            23832,
  "        ",      13823,
  "      ",         31415,
  "      ",         28461,
  "      ",        20645,
  "     ",          20900,
  "      ",         8097,
  "       ",       31865
) |>
  mutate(total_ha = total_km2 * 100)

sown_pct <- sown |>
  left_join(oblast_area_km2 |> dplyr::select(region, total_ha), by = "region") |>
  mutate(sown_pct = sown_ha / total_ha * 100) |>
  filter(!is.na(sown_pct))

p_sownpct_west <- plot_small_multiple(
  sown_pct, region_groups, "sown_pct", "West",
  title    = "Sown wheat area as % of total oblast area - West",
  subtitle = "% of total oblast land area planted with winter wheat, 2015-2025.",
  caption  = "Oblast areas: official administrative data. Sown area: SSU."
)

p_sownpct_centre <- plot_small_multiple(
  sown_pct, region_groups, "sown_pct", "Centre",
  title    = "Sown wheat area as % of total oblast area - Centre",
  subtitle = "% of total oblast land area planted with winter wheat, 2015-2025.",
  caption  = "Oblast areas: official administrative data. Sown area: SSU."
)

p_sownpct_east <- plot_small_multiple(
  sown_pct, region_groups, "sown_pct", "East",
  title    = "Sown wheat area as % of total oblast area - East",
  subtitle = "% of total oblast land area planted with winter wheat, 2015-2025.",
  caption  = "Oblast areas: official administrative data. Sown area: SSU."
)
```

```r
sown_pct_avg <- sown_pct |>
  left_join(region_groups, by = "region") |>
  group_by(region, group) |>
  summarise(mean_pct = mean(sown_pct, na.rm = TRUE), .groups = "drop") |>
  arrange(mean_pct) |>
  mutate(region = factor(region, levels = region))

p_sownpct_bar <- ggplot(sown_pct_avg,
                        aes(x = mean_pct, y = region, fill = group)) +
  geom_col(width = 0.72, alpha = 0.88) +
  geom_text(aes(label = paste0(round(mean_pct, 1), "%")),
            hjust = -0.1, size = 2.6, colour = "grey30") +
  scale_fill_manual(values = GROUP_COLORS) +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.12)),
    labels = scales::label_number(suffix = "%")
  ) +
  labs(
    title    = "Average sown wheat area as % of total oblast area (2015-2025)",
    subtitle = "Oblasts ordered by intensity. Colour = geographical group.",
    caption  = "Oblast areas: official administrative data. Sown area: SSU.",
    fill     = "Group"
  ) +
  theme_small_multiple(base_size = 10) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.major.x = element_line(colour = "grey90", linewidth = 0.3),
    axis.line.x        = element_blank(),
    axis.text.y        = element_text(size = 8),
    legend.position    = "right",
    legend.text        = element_text(size = 8),
    legend.title       = element_text(size = 8, face = "bold")
  )


rep_oblasts <- yield |>
  left_join(region_groups, by = "region") |>
  filter(!is.na(group), !is.na(yield_cpha)) |>
  group_by(group, region) |>
  summarise(mean_y = mean(yield_cpha, na.rm = TRUE), .groups = "drop") |>
  slice_max(mean_y, n = 1, by = group)

rep_data <- yield |>
  inner_join(rep_oblasts |> dplyr::select(group, region), by = "region") |>
  filter(!is.na(yield_cpha))

p_representative <- ggplot() +
  geom_line(
    data    = ukr_yield |> filter(!is.na(yield_cpha)),
    mapping = aes(x = year, y = yield_cpha),
    colour  = "grey65", linewidth = 0.7, linetype = "dashed"
  ) +
  geom_line(
```

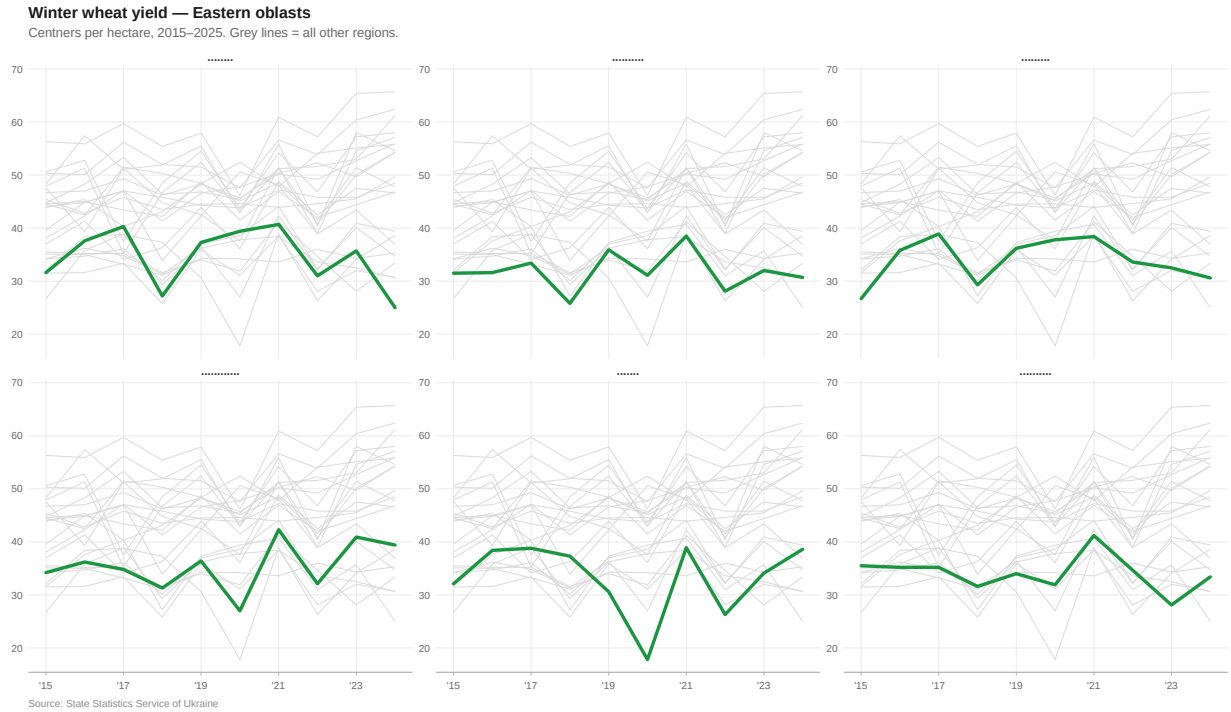
```r
    data    = rep_data,
    mapping = aes(x = year, y = yield_cpha, colour = group),
    linewidth = 1.2, lineend = "round"
  ) +
  facet_wrap(~group) +
  scale_colour_manual(values = GROUP_COLORS) +
  scale_x_continuous(
    breaks = c(2015, 2017, 2019, 2021, 2023, 2025),
    labels = c("'15", "'17", "'19", "'21", "'23", "'25")
  ) +
  labs(
    title    = "Most productive representative oblast per geographical group",
    subtitle = "Dashed grey = Ukraine national average. Coloured = representative oblast (highest mean y
    caption  = "Source: State Statistics Service of Ukraine"
  ) +
  theme_small_multiple(base_size = 10) +
  theme(strip.text = element_text(face = "bold", size = 10, colour = "grey15"))


print(p_yield_west)
```



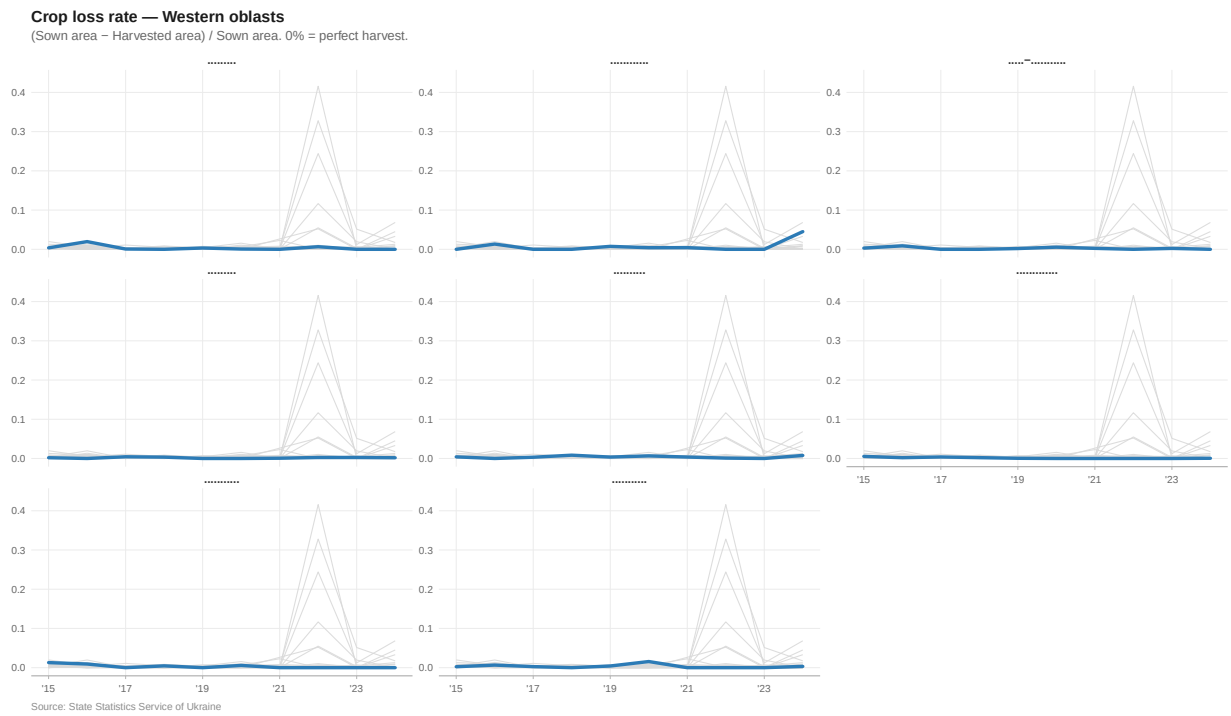

```r
print(p_yield_centre)
```



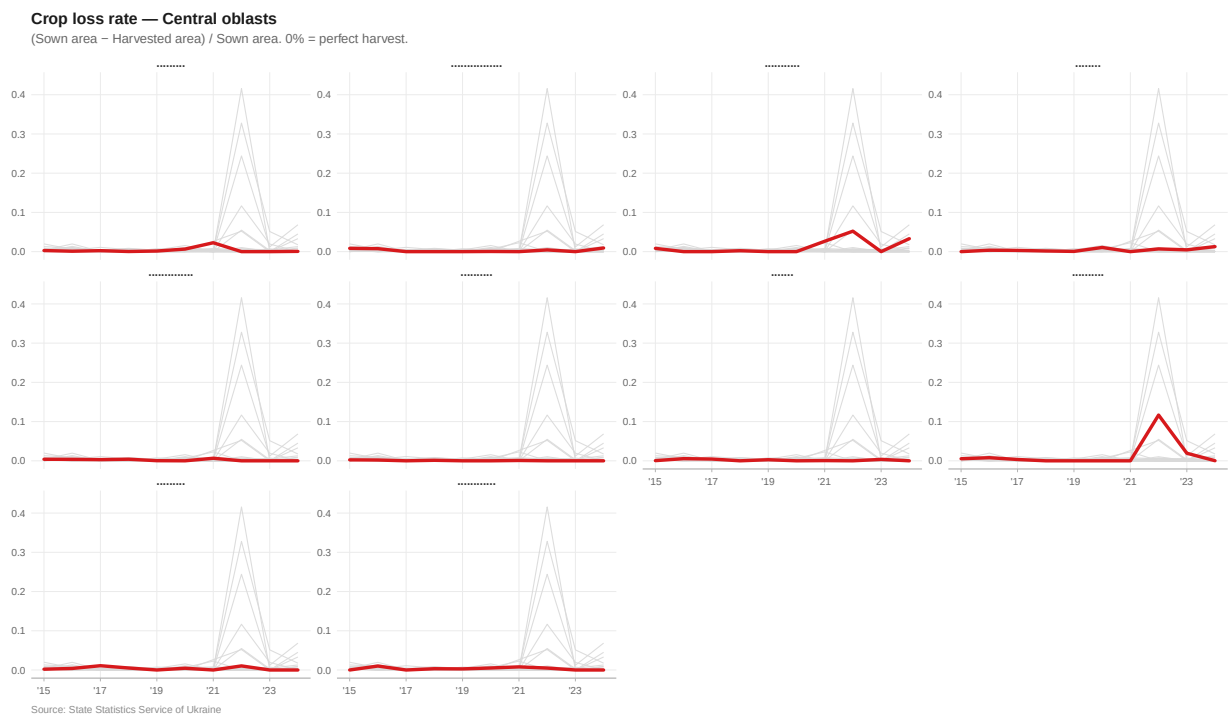

```
print(p_yield_east)
```



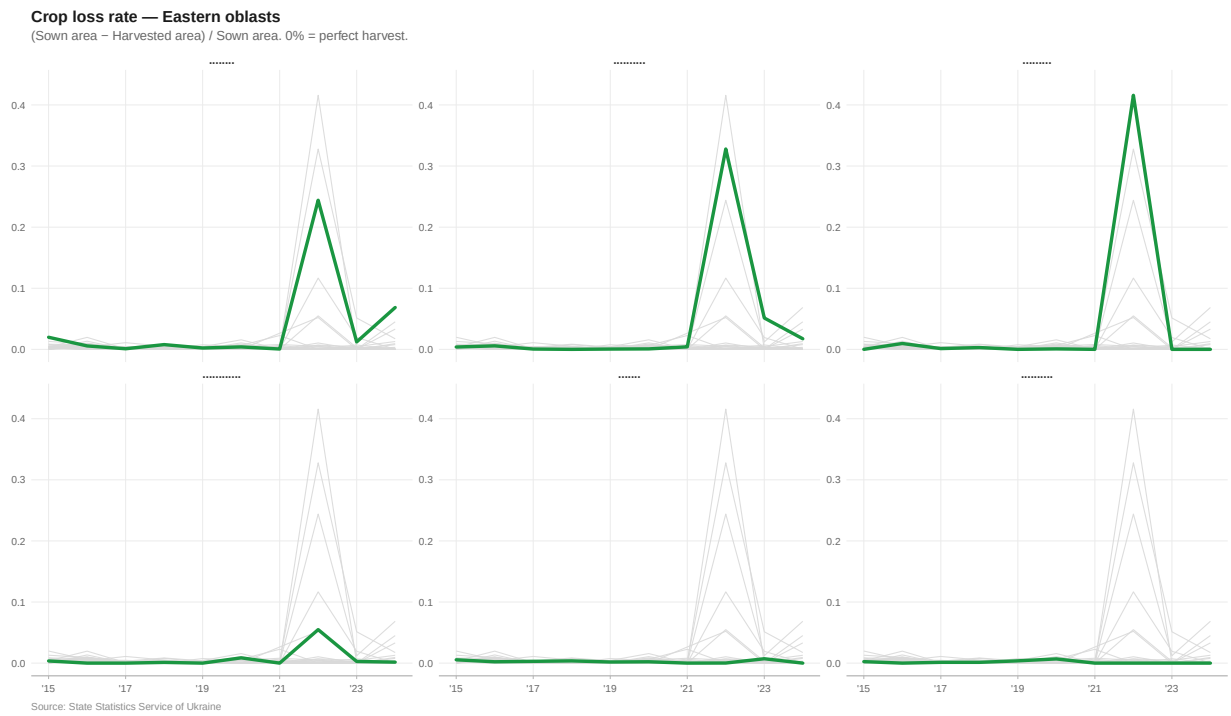

```
print(p_loss_west)
```



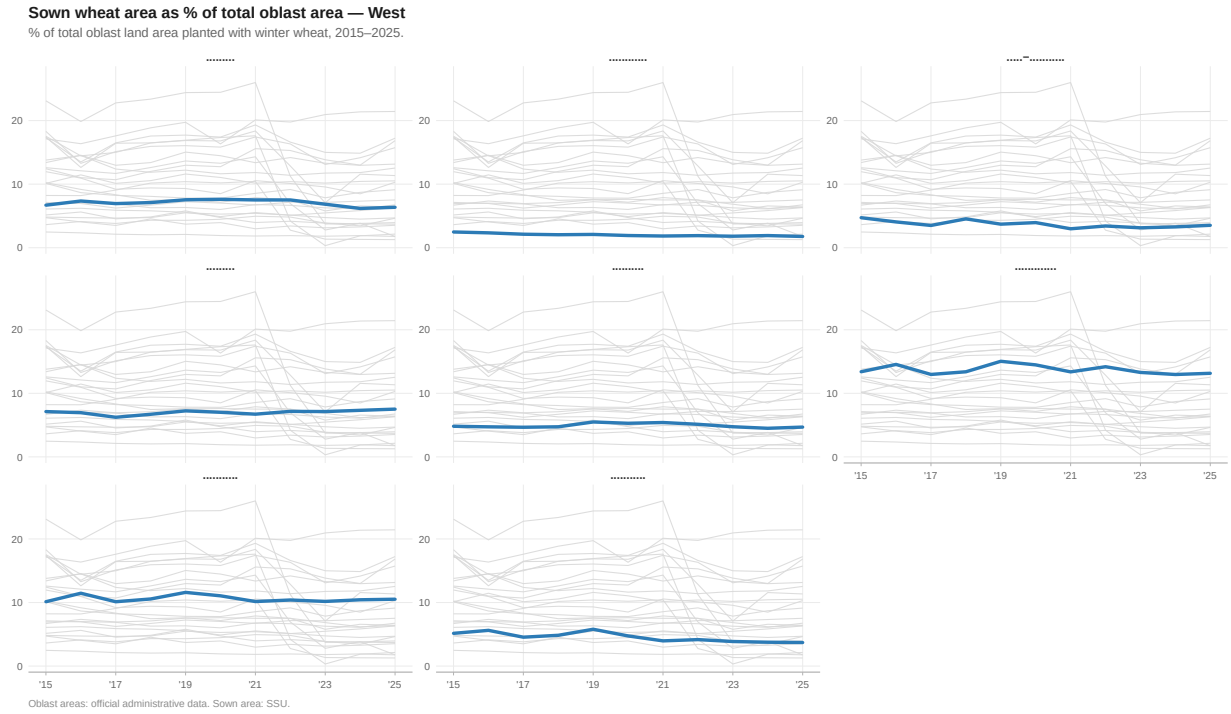

```
print(p_loss_centre)
```



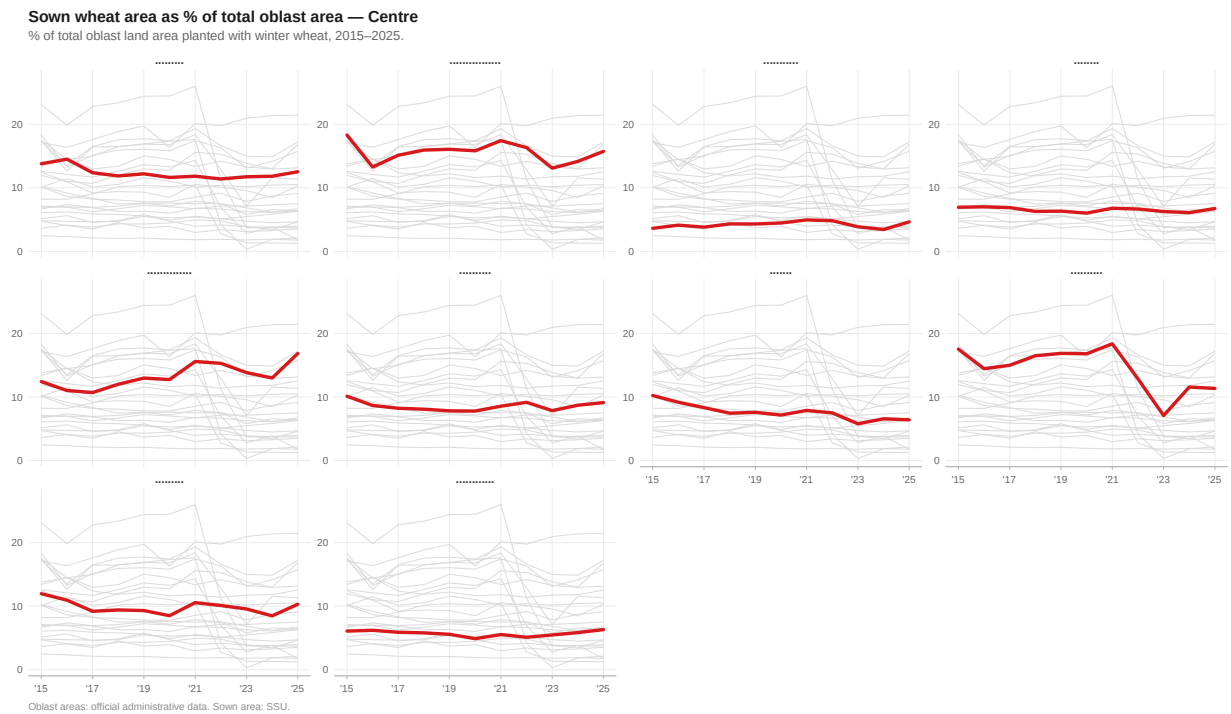

```
print(p_loss_east)
```



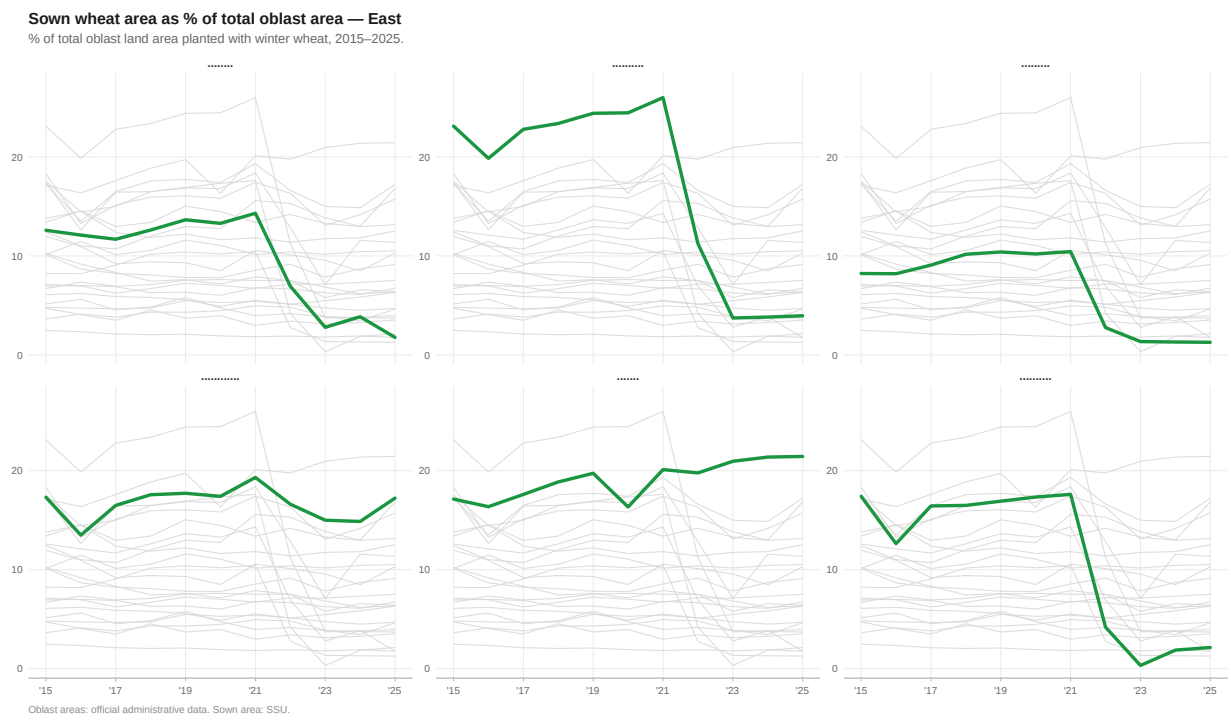

```
print(p_sownpct_west)
```



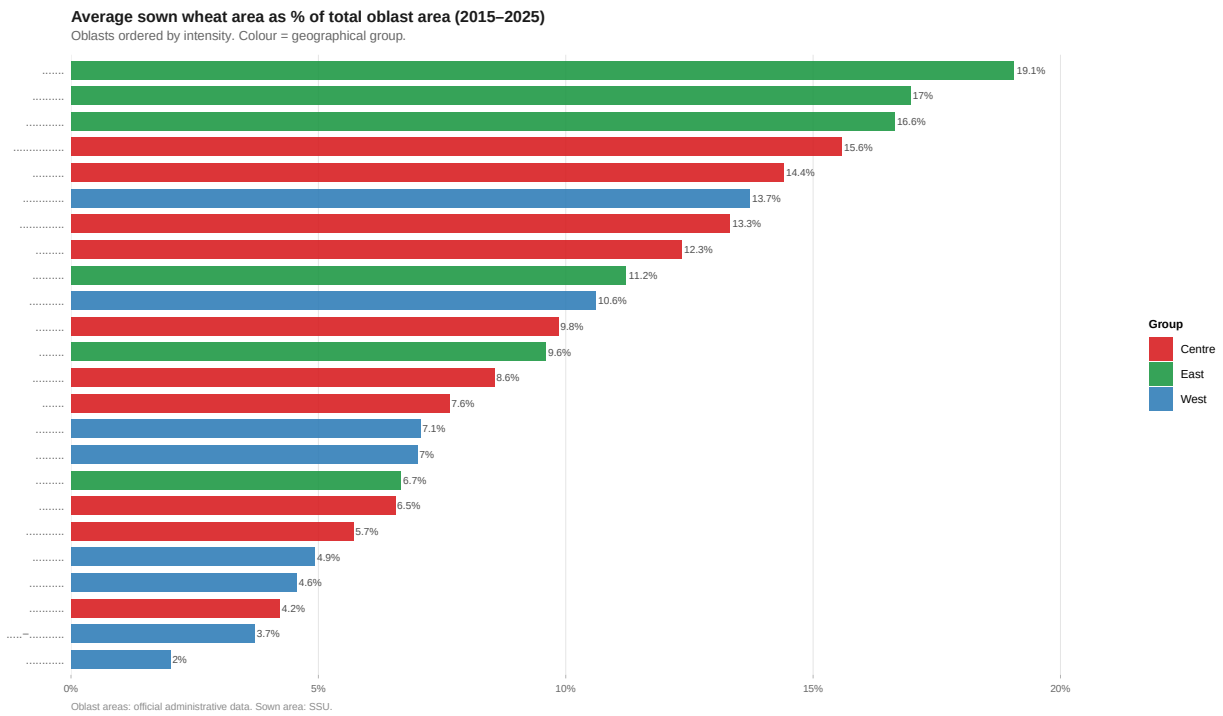

```
print(p_sownpct_centre)
```



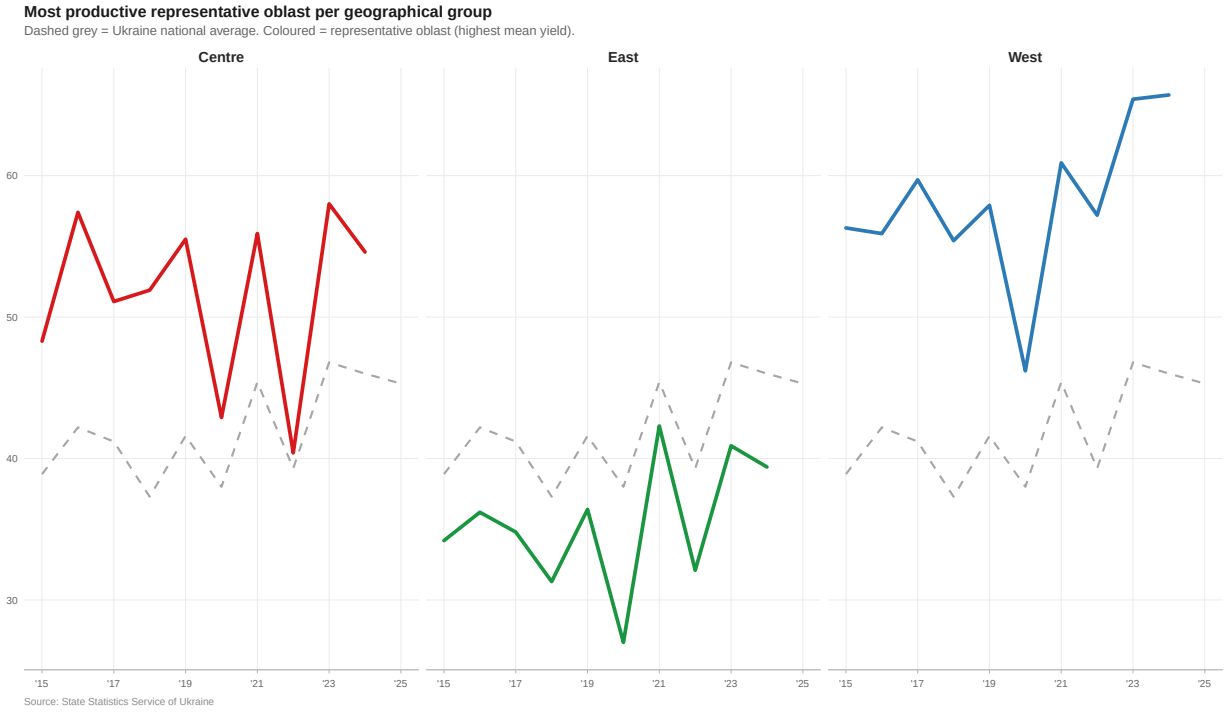

```
print(p_sownpct_east)
```




```
print(p_sownpct_bar)
```




```
print(p_representative)
```



Yield Patterns and Regional Heterogeneity The small-multiple yield plots reveal substantial variation across oblasts both in levels and in temporal dynamics. Western regions (Vinnytsia, Khmelnytskyi, Ternopil) consistently show higher yields (45–75 ц/га) with relatively

stable trajectories, while eastern oblasts (Odesa, Zaporizhzhia, Donetsk) exhibit lower and more volatile yields (25–55 ц/га), likely reflecting differences in soil quality, moisture availability, and war-related disruptions after 2022.

Critically, the plots demonstrate that **different regions follow different yield trends**. Some oblasts show gradual increases over 2015–2021 (suggesting technology adoption or management improvements), others remain flat, and a few (particularly in the east) display declining pre-war trends. This heterogeneity has direct implications for our modelling strategy: a single common time trend will be insufficient. We will therefore include both a common trend (capturing Ukraine-wide technological progress) and region-specific trends (allowing each oblast's trajectory to diverge).

Loss Rates and the Lagged Effect Argument The loss rate plots — defined as `(sown area - harvested area) / sown area` — show that even in normal years, 2–8% of sown wheat area is lost before harvest. In extreme years (e.g., 2020 in Odesa, 2022 in eastern oblasts), losses exceed 15–20%.

Why does this matter for modelling current-year yield? **Losses in year t−1 can affect yield in year t through farmer behavioural responses.** After a year of high losses, farmers may:

1. **Increase input intensity** (more fertiliser, better seeds, more frequent fungicide applications) to compensate for lost income or to secure the next harvest
2. **Alter sowing decisions** (switch to more resilient varieties, adjust planting dates, change crop rotation)
3. **Face financial constraints** that limit their ability to invest in the following season

Conversely, low-loss years may create complacency or reduce the perceived need for intensive management. These dynamics imply that the loss rate from the previous year — which we denote as `loss_lag` — is a legitimate predictor of current yield, capturing unobserved farmer responses that are not directly measured in our data. We will test this hypothesis explicitly in the modelling section by including `loss_lag` as a regressor.

2.2 Daily Weather Data (`daily_weather.csv`) and Soil Moisture

Variable	Description	Unit
`region`	Oblast name	text
`date`	Calendar date	YYYY-MM-DD
`temperature_2m_max`	Daily maximum temperature	°C
`temperature_2m_min`	Daily minimum temperature	°C
`temperature_2m_mean`	Daily mean temperature	°C
`wind_speed_10m_max`	Daily maximum wind speed at 10m height	m/s
`wind_gusts_10m_max`	Daily maximum wind gusts at 10m height	m/s
`et0_fao_evapotranspiration`	Reference evapotranspiration (FAO Penman-Monteith)	mm/day
`dewpoint_2m_mean`	Daily mean dewpoint temperature	°C

Variable	Description	Unit
`precipitation_sum`	Daily total precipitation	mm
`shortwave_radiation_sum`	Daily incoming shortwave solar radiation	MJ/m²

Additional data — soil moisture (`soil_moisture.csv`):

Variable	Description	Unit
`region`	Oblast name	text
`date`	Calendar date	YYYY-MM-DD
`soil_moisture_0_7cm_pct`	Volumetric soil moisture (0–7 cm depth)	%

Notes: ET_0 represents atmospheric evaporative demand — higher values indicate faster water loss from crops. Dewpoint is a direct measure of air moisture (higher values promote fungal diseases). Data cover 2013–2024 for all 24 oblasts. Soil moisture data are sourced from Open-Meteo ERA5-Land archive.

Expected Correlations Among Weather Parameters Understanding correlations helps us avoid multicollinearity in later models:

- **ET_0, solar radiation, and temperature** — strongly positively correlated. We will select only ET_0 as the representative.
- **Precipitation and soil moisture** — moderately correlated (1–3 day lag), but soil moisture captures drought persistence uniquely.
- **Dewpoint and temperature** — context-dependent correlation; dewpoint carries disease-pressure information.
- **Wind speed** — largely uncorrelated with others; kept for lodging risk.
- **Soil moisture vs. temperature/ET_0** — negative correlation (hot conditions deplete soil water).

Constructed Weather Metrics (following the research hypothesis) Why aggregate daily data? Our hypothesis posits that **extreme days matter more than averages**. Using daily data directly would create thousands of predictors. The solution is to aggregate into theoretically grounded stress metrics, preserving non-linear "extreme shock" logic while collapsing 365 days into few annual numbers.

Information loss recovery: Aggregation loses timing. We recover this by splitting the year into meaningful periods (winter, early spring, late spring) and calculating metrics separately for each window.

For each region and year, **temperature** is transformed as:

$$t_{dev} = \frac{T_{max} + T_{min}}{2} - 5$$

where 5°C is the assumed base temperature for winter wheat development.

From this deviation, four temperature metrics are calculated:

Metric	Definition
`days_hot_critical`	Number of days with $t_{dev} > 20$ (extreme heat stress)
`days_cold_critical`	Number of days with $t_{dev} < -15$ (extreme cold stress)
`sum_heat_noncrit`	Sum of t_{dev} for days with $0 < t_{dev} \leq 20$
`sum_cold_noncrit`	Sum of t_{dev} for days with $-15 \leq t_{dev} < 0$

For **soil moisture**, we apply the **exact same logic** — counting extreme events rather than using raw values:

Metric	Definition
`days_drought`	Number of days with soil moisture < 20% (severe water stress)
`days_waterlog`	Number of days with soil moisture > 85% (root asphyxiation risk)
`mean_soil_moisture`	Annual average soil moisture (baseline condition)

Why these thresholds? For winter wheat: - **<20% soil moisture** — permanent wilting point, stomata close, photosynthesis stops - **>85% soil moisture** — oxygen depletion in root zone, root rot and fungal diseases

This mirrors our temperature logic: just as we count `days_hot_crit` and `days_cold_crit`, we now count `days_drought` and `days_waterlog` as the water-side equivalents.

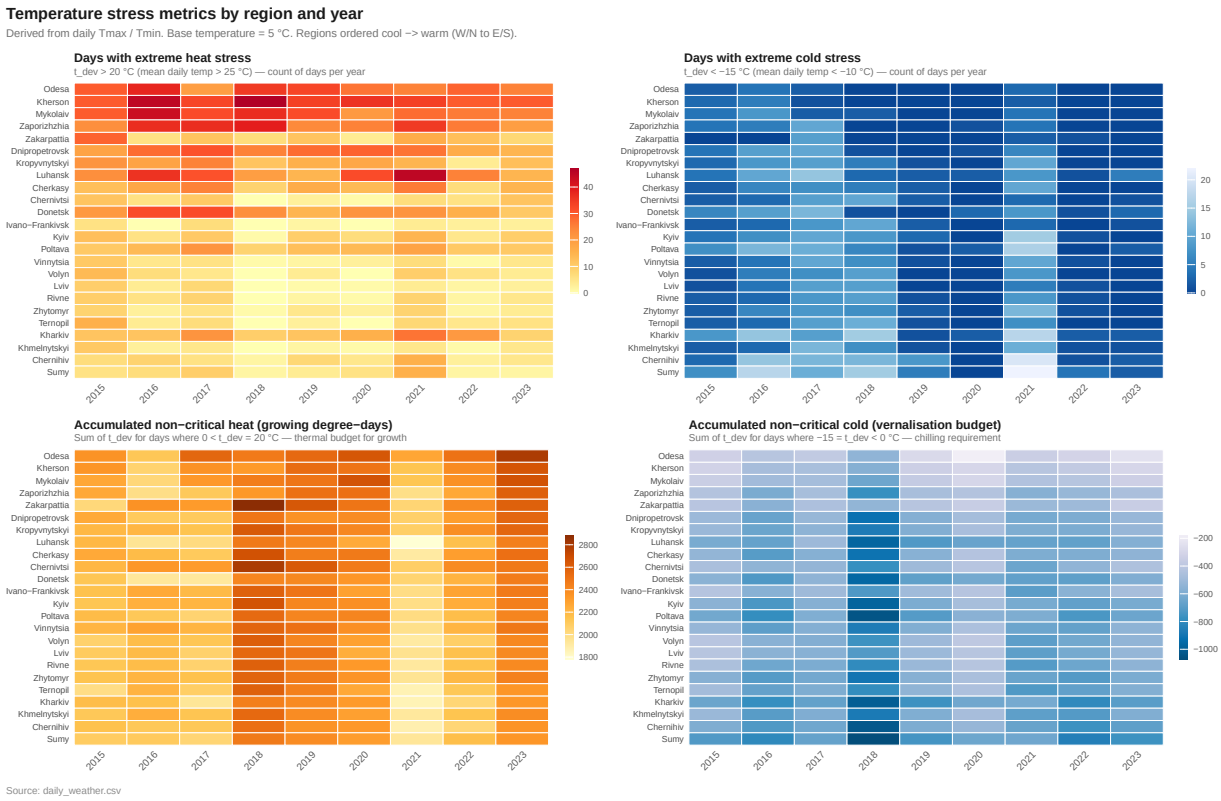
Note on trends: All weather metrics exhibit trends over time (warming climate). These will be absorbed by common time trends (`t`) and region-specific trends (`region × t`) in the modelling section.

```r
#   Load and aggregate weather data
weather <- read_csv("daily_weather.csv", show_col_types = FALSE) |>
  mutate(date = as_date(date), year = year(date)) |>
  filter(year >= 2013, year <= 2024)

#   Load and aggregate soil moisture data
# Same logic as temperature: count extreme days, not just averages
soil_moisture <- read_csv("soil_moisture.csv", show_col_types = FALSE) |>
  mutate(date = as_date(date), year = year(date)) |>
  filter(year >= 2013, year <= 2024)

soil_moisture_agg <- soil_moisture |>
  mutate(
    # Same extreme-day counting logic as temperature
    drought_day = soil_moisture_0_7cm_pct < 20,
    waterlog_day = soil_moisture_0_7cm_pct > 85
  ) |>
  group_by(region, year) |>
  summarise(
    mean_soil_moisture_pct = mean(soil_moisture_0_7cm_pct, na.rm = TRUE),
    days_drought = sum(drought_day, na.rm = TRUE),
    days_waterlog = sum(waterlog_day, na.rm = TRUE),
```

```r
    .groups = "drop"
  )

BASE_TEMP <- 5

weather_annual <- weather |>
  mutate(
    t_mean         = (temperature_2m_max + temperature_2m_min) / 2,
    t_dev          = t_mean - BASE_TEMP,
    days_hot_crit  = t_dev > 20,
    days_cold_crit = t_dev < -15,
    heat_noncrit   = if_else(t_dev > 0  & t_dev <= 20,  t_dev, 0),
    cold_noncrit   = if_else(t_dev >= -15 & t_dev < 0,  t_dev, 0)
  ) |>
  group_by(region, year) |>
  summarise(
    # Thermal stress metrics
    days_hot_crit    = sum(days_hot_crit,  na.rm = TRUE),
    days_cold_crit   = sum(days_cold_crit, na.rm = TRUE),
    sum_heat_noncrit = sum(heat_noncrit,   na.rm = TRUE),
    sum_cold_noncrit = sum(cold_noncrit,   na.rm = TRUE),
    # Raw temperature levels
    mean_tmax        = mean(temperature_2m_max,          na.rm = TRUE),
    mean_tmin        = mean(temperature_2m_min,          na.rm = TRUE),
    mean_temp        = mean(t_mean,                      na.rm = TRUE),
    # Water and energy
    total_precip     = sum(precipitation_sum,            na.rm = TRUE),
    total_et0        = sum(et0_fao_evapotranspiration,   na.rm = TRUE),
    total_radiation  = sum(shortwave_radiation_sum,      na.rm = TRUE),
    mean_wind        = mean(wind_gusts_10m_max,          na.rm = TRUE),
    mean_dewpoint    = mean(dewpoint_2m_mean,            na.rm = TRUE),
    .groups = "drop"
  ) |>
  left_join(soil_moisture_agg, by = c("region", "year"))

# Order regions W→E by mean temperature
region_order <- weather_annual |>
  group_by(region) |>
  summarise(mt = mean(mean_temp, na.rm = TRUE)) |>
  arrange(mt) |>
  pull(region)

weather_annual <- weather_annual |>
  mutate(region = factor(region, levels = region_order),
         year   = factor(year))

#   Shared theme
theme_heatmap <- function(base_size = 9) {
  theme_minimal(base_size = base_size) +
    theme(
      panel.grid        = element_blank(),
      panel.background  = element_rect(fill = "white", colour = NA),
      plot.background   = element_rect(fill = "white", colour = NA),
```

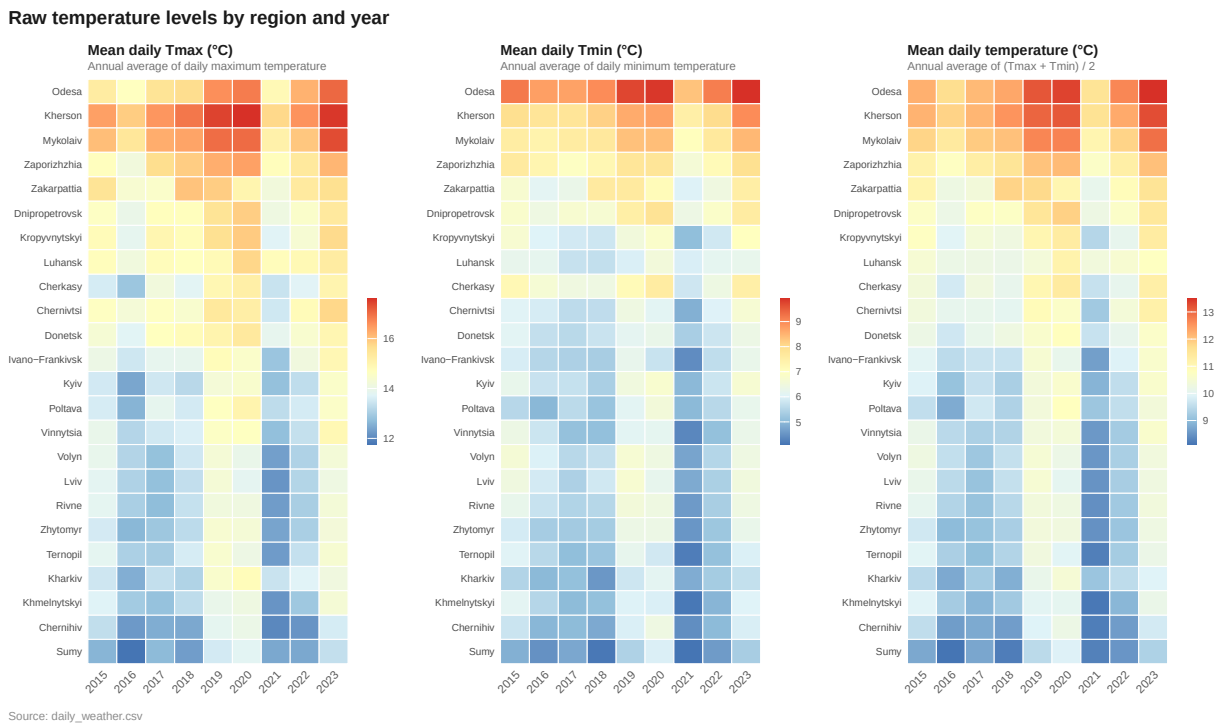
```r
      axis.text.x       = element_text(size = 7.5, angle = 45, hjust = 1,
                                       colour = "grey35"),
      axis.text.y       = element_text(size = 7,   colour = "grey30"),
      axis.title        = element_blank(),
      axis.ticks        = element_blank(),
      strip.text        = element_text(face = "bold", size = 8.5,
                                       colour = "grey15"),
      strip.background  = element_blank(),
      plot.title        = element_text(face = "bold", size = 10,
                                       colour = "grey10",
                                       margin = margin(b = 2)),
      plot.subtitle     = element_text(size = 7.5, colour = "grey45",
                                       margin = margin(b = 5)),
      legend.text       = element_text(size = 7,   colour = "grey35"),
      legend.title      = element_blank(),
      legend.key.width  = unit(0.4, "cm"),
      legend.key.height = unit(1.5, "cm"),
      plot.margin       = margin(6, 8, 6, 6)
    )
}

#   Helper: one heatmap tile
make_tile <- function(df, metric, palette, direction = 1,
                      title, subtitle) {
  ggplot(df, aes(x = year, y = region, fill = .data[[metric]])) +
    geom_tile(colour = "white", linewidth = 0.35) +
    scale_fill_distiller(palette   = palette,
                         direction = direction,
                         guide     = guide_colorbar(
                           barwidth  = 0.45,
                           barheight = 7,
                           ticks     = FALSE)) +
    scale_x_discrete(expand = c(0, 0)) +
    scale_y_discrete(expand = c(0, 0)) +
    labs(title = title, subtitle = subtitle) +
    theme_heatmap()
}

#   Temperature stress metrics
p1 <- make_tile(weather_annual, "days_hot_crit", "YlOrRd", 1,
  "Days with extreme heat stress",
  "t_dev > 20 °C (mean daily temp > 25 °C) - count of days per year")

p2 <- make_tile(weather_annual, "days_cold_crit", "Blues", -1,
  "Days with extreme cold stress",
  "t_dev < -15 °C (mean daily temp < -10 °C) - count of days per year")

p3 <- make_tile(weather_annual, "sum_heat_noncrit", "YlOrBr", 1,
  "Accumulated non-critical heat (growing degree-days)",
  "Sum of t_dev for days where 0 < t_dev   20 °C - thermal budget for growth")

p4 <- make_tile(weather_annual, "sum_cold_noncrit", "PuBu", -1,
  "Accumulated non-critical cold (vernalisation budget)",
```

```
  "Sum of t_dev for days where -15   t_dev < 0 °C - chilling requirement")

(p1 | p2) / (p3 | p4) +
  plot_annotation(
    title    = "Temperature stress metrics by region and year",
    subtitle = paste0("Derived from daily Tmax / Tmin. Base temperature = ",
                      BASE_TEMP, " °C. Regions ordered cool → warm (W/N to E/S)."),
    caption  = "Source: daily_weather.csv",
    theme    = theme(
      plot.title      = element_text(face = "bold", size = 13, colour = "grey10"),
      plot.subtitle   = element_text(size = 9, colour = "grey45"),
      plot.caption    = element_text(size = 7.5, colour = "grey55", hjust = 0),
      plot.background = element_rect(fill = "white", colour = NA)
    )
  )
```

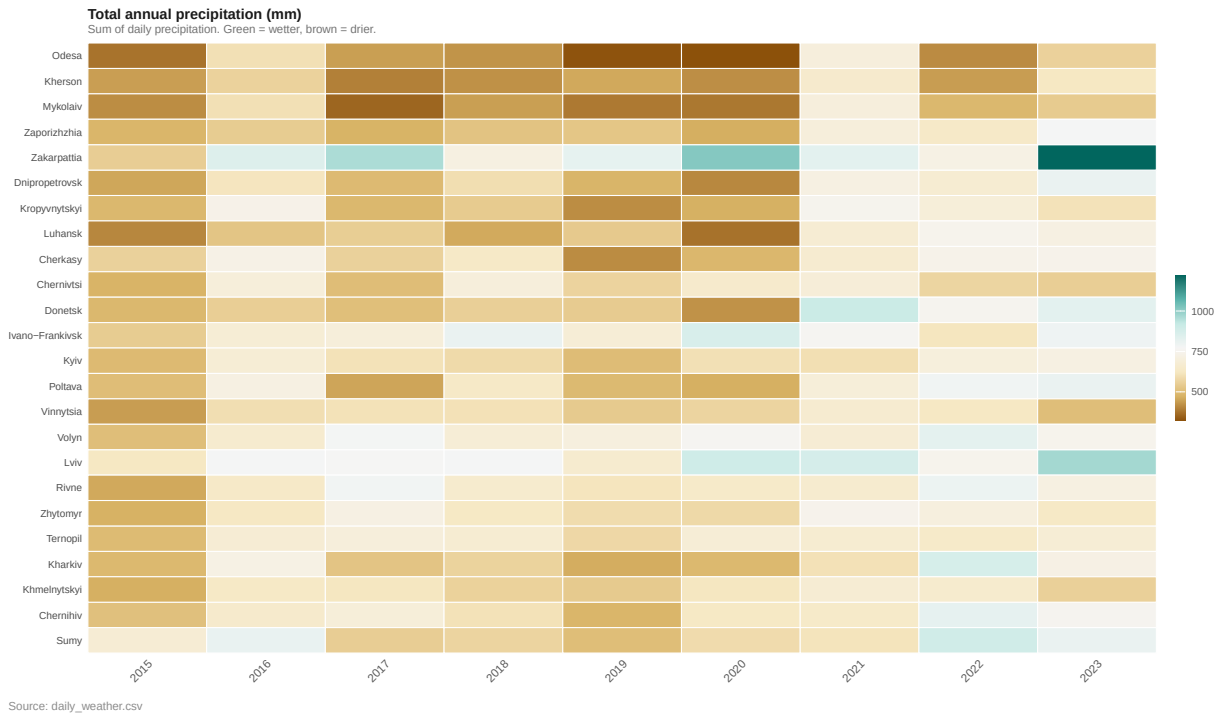



```
#   Raw temperature levels
p_tmax <- make_tile(weather_annual, "mean_tmax", "RdYlBu", -1,
  "Mean daily Tmax (°C)",
  "Annual average of daily maximum temperature")

p_tmin <- make_tile(weather_annual, "mean_tmin", "RdYlBu", -1,
  "Mean daily Tmin (°C)",
  "Annual average of daily minimum temperature")

p_tmean <- make_tile(weather_annual, "mean_temp", "RdYlBu", -1,
  "Mean daily temperature (°C)",
```

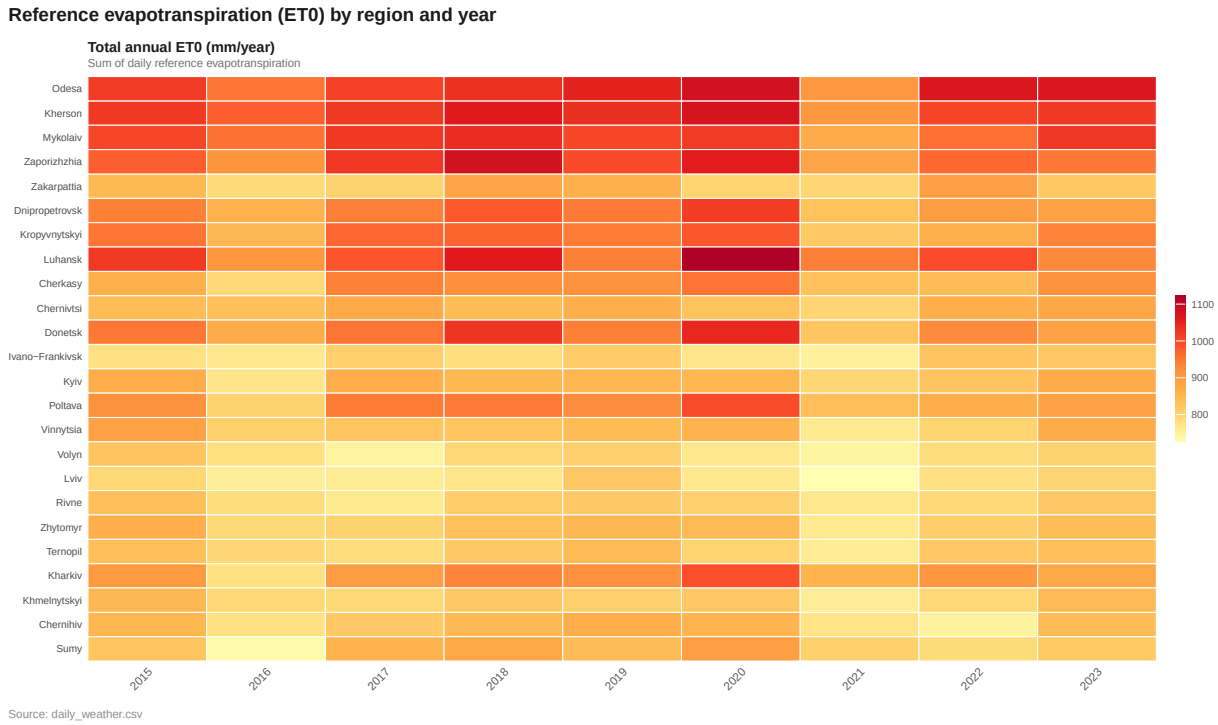
```r
  "Annual average of (Tmax + Tmin) / 2")

(p_tmax | p_tmin | p_tmean) +
  plot_annotation(
    title   = "Raw temperature levels by region and year",
    caption = "Source: daily_weather.csv",
    theme   = theme(
      plot.title      = element_text(face = "bold", size = 13, colour = "grey10"),
      plot.caption    = element_text(size = 7.5, colour = "grey55", hjust = 0),
      plot.background = element_rect(fill = "white", colour = NA)
    )
  )
```



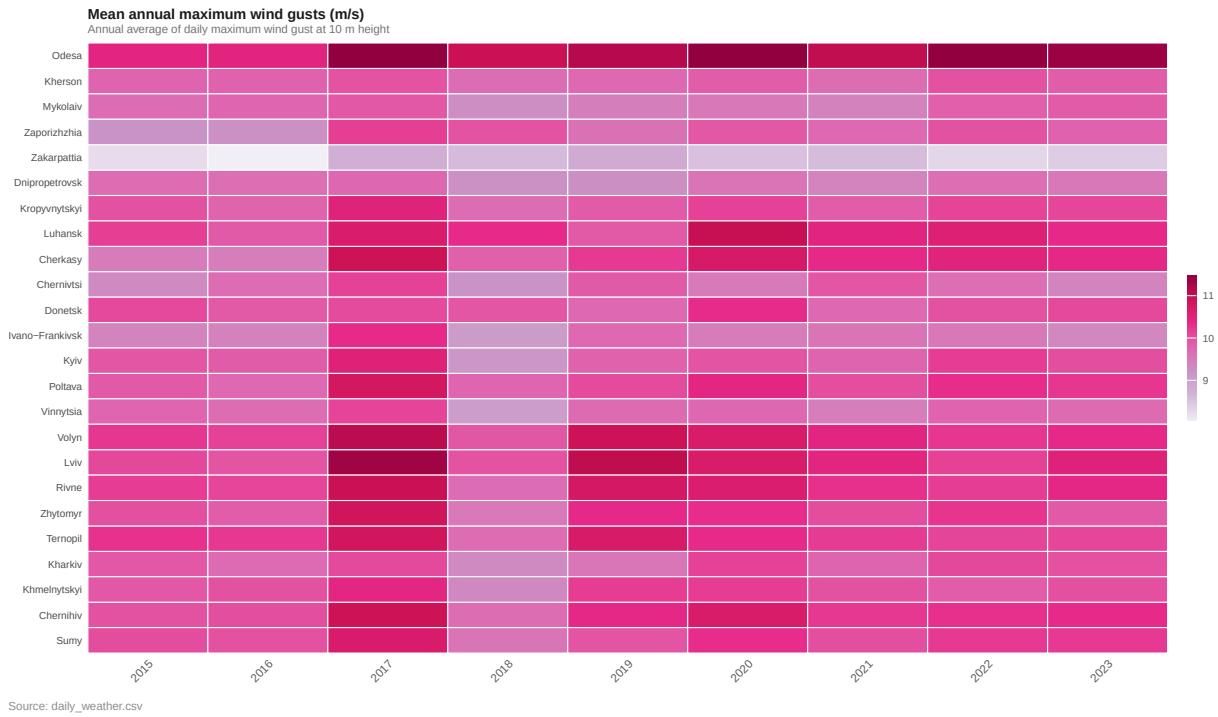

```r
#   Precipitation
make_tile(weather_annual, "total_precip", "BrBG", 1,
  "Total annual precipitation (mm)",
  "Sum of daily precipitation. Green = wetter, brown = drier.") +
  plot_annotation(
    caption = "Source: daily_weather.csv",
    theme   = theme(
      plot.background = element_rect(fill = "white", colour = NA),
      plot.caption    = element_text(size = 7.5, colour = "grey55", hjust = 0)
    )
  )
```



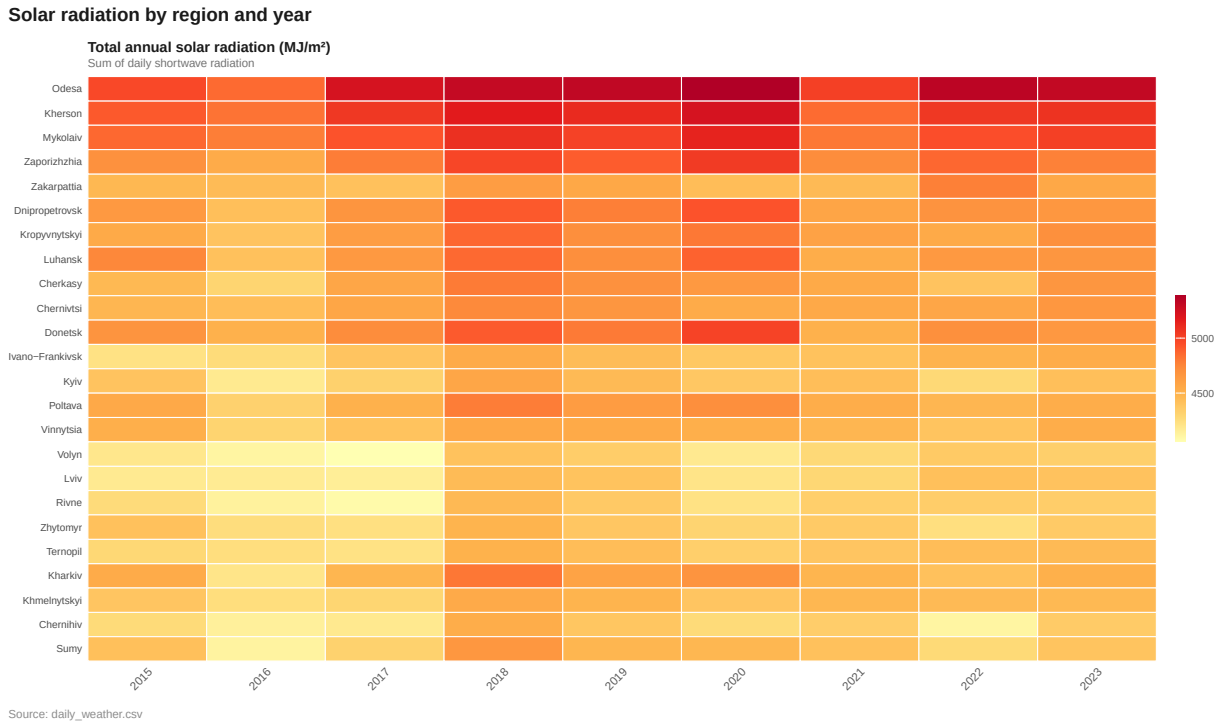

```r
#   Evapotranspiration
make_tile(weather_annual, "total_et0", "YlOrRd", 1,
  "Total annual ET  (mm/year)",
  "Sum of daily reference evapotranspiration") +
  plot_annotation(
    title   = "Reference evapotranspiration (ET ) by region and year",
    caption = "Source: daily_weather.csv",
    theme   = theme(
      plot.title      = element_text(face = "bold", size = 13, colour = "grey10"),
      plot.caption    = element_text(size = 7.5, colour = "grey55", hjust = 0),
      plot.background = element_rect(fill = "white", colour = NA)
    )
  )
```



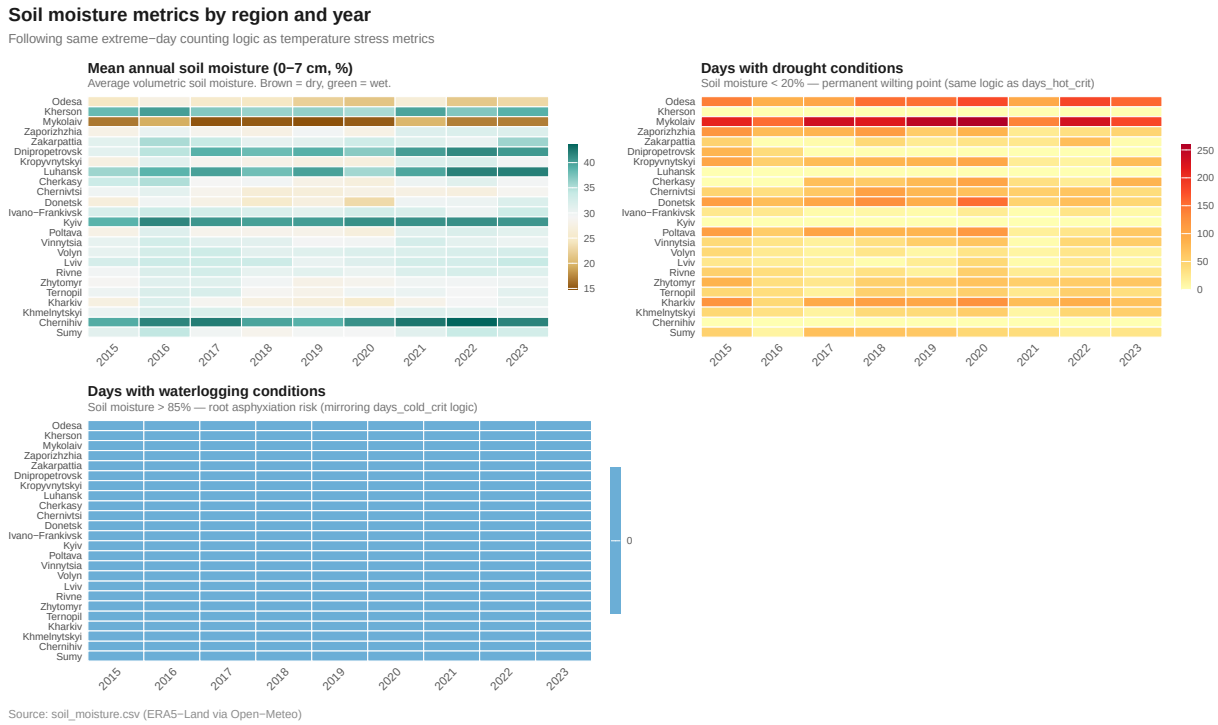

```r
#   Wind speed
make_tile(weather_annual, "mean_wind", "PuRd", 1,
  "Mean annual maximum wind gusts (m/s)",
  "Annual average of daily maximum wind gust at 10 m height") +
  plot_annotation(
    caption = "Source: daily_weather.csv",
    theme   = theme(
      plot.background = element_rect(fill = "white", colour = NA),
      plot.caption    = element_text(size = 7.5, colour = "grey55", hjust = 0)
    )
  )
```




```r
#   Solar radiation
make_tile(weather_annual, "total_radiation", "YlOrRd", 1,
  "Total annual solar radiation (MJ/m²)",
  "Sum of daily shortwave radiation") +
  plot_annotation(
    title   = "Solar radiation by region and year",
    caption = "Source: daily_weather.csv",
    theme   = theme(
      plot.title      = element_text(face = "bold", size = 13, colour = "grey10"),
      plot.caption    = element_text(size = 7.5, colour = "grey55", hjust = 0),
      plot.background = element_rect(fill = "white", colour = NA)
    )
  )
```




```r
#   Soil moisture metrics (same logic as temperature)
p_soil_mean <- make_tile(weather_annual, "mean_soil_moisture_pct", "BrBG", 1,
  "Mean annual soil moisture (0-7 cm, %)",
  "Average volumetric soil moisture. Brown = dry, green = wet.")

p_drought <- make_tile(weather_annual, "days_drought", "YlOrRd", 1,
  "Days with drought conditions",
  "Soil moisture < 20% - permanent wilting point (same logic as days_hot_crit)")

p_waterlog <- make_tile(weather_annual, "days_waterlog", "Blues", -1,
  "Days with waterlogging conditions",
  "Soil moisture > 85% - root asphyxiation risk (mirroring days_cold_crit logic)")

(p_soil_mean | p_drought) / (p_waterlog | plot_spacer()) +
  plot_annotation(
    title   = "Soil moisture metrics by region and year",
    subtitle = "Following same extreme-day counting logic as temperature stress metrics",
    caption = "Source: soil_moisture.csv (ERA5-Land via Open-Meteo)",
    theme   = theme(
      plot.title      = element_text(face = "bold", size = 13, colour = "grey10"),
      plot.subtitle   = element_text(size = 9, colour = "grey45"),
      plot.caption    = element_text(size = 7.5, colour = "grey55", hjust = 0),
      plot.background = element_rect(fill = "white", colour = NA)
    )
  )
```



Note on waterlogging detection: The soil moisture heatmaps show virtually zero days exceeding the 85% waterlogging threshold across all regions and years. This does not imply that floods or waterlogging events did not occur in Ukraine during 2015–2023. Rather, this reflects two limitations of our data: first, the ERA5-Land reanalysis product smooths extreme moisture events, potentially underestimating short-term saturation episodes; second, the 0–7 cm surface layer drains rapidly and does not capture waterlogging in the deeper root zone (20–60 cm), where crop damage actually occurs. Consequently, our days_waterlog metric should be interpreted as a measure of extreme, persistent surface saturation rather than a comprehensive indicator of flood risk. For practical agronomic applications, precipitation intensity (e.g., daily rainfall >40 mm) or multi-day cumulative rainfall may serve as more reliable proxies for waterlogging risk.

```r
#   Summary table
weather_annual |>
  group_by(region) |>
  summarise(
    `Hot days`      = round(mean(days_hot_crit, na.rm = TRUE),     1),
    `Cold days`     = round(mean(days_cold_crit, na.rm = TRUE),    1),
    `Heat sum`      = round(mean(sum_heat_noncrit, na.rm = TRUE), 0),
    `Cold sum`      = round(mean(sum_cold_noncrit, na.rm = TRUE), 0),
    `Precip (mm)`   = round(mean(total_precip, na.rm = TRUE),      0),
    `ET  (mm/yr)`   = round(mean(total_et0, na.rm = TRUE),         0),
    `Wind (m/s)`    = round(mean(mean_wind, na.rm = TRUE),         1),
    `Soil moisture (%)` = round(mean(mean_soil_moisture_pct, na.rm = TRUE), 1),
    `Drought days` = round(mean(days_drought, na.rm = TRUE),      1),
    `Waterlog days` = round(mean(days_waterlog, na.rm = TRUE),     1)
  ) |>
  arrange(desc(`Hot days`)) |>
```

```
knitr::kable(caption = "Mean annual weather and soil moisture metrics by oblast (2013-2024, ordered by
```

Table 6: Mean annual weather and soil moisture metrics by oblast (2013–2024, ordered by heat stress days)

region	Hot days	Cold days	Heat sum	Cold sum	Precip (mm)	ET_0 (mm/yr)	Wind (m/s)	Soil moisture (%)	Drought days	Waterlog days
Kherson	35.4	1.2	2369	-397	490	1014	9.8	37.7	0.0	0
Mykolaiv	30.4	2.2	2381	-427	478	988	9.6	16.7	209.0	0
Zaporizhzhia	29.6	2.7	2288	-528	568	982	9.7	29.7	71.0	0
Odesa	29.1	1.2	2476	-339	462	1020	11.1	24.0	139.2	0
Luhansk	26.9	5.4	2188	-674	553	988	10.4	39.1	0.0	0
Dnipropetrovsk	24.3	3.4	2296	-601	581	922	9.5	37.9	13.7	0
Donetsk	21.6	4.8	2212	-673	616	936	10.0	28.0	91.8	0
Cherkasy	16.0	3.6	2321	-608	612	886	10.2	30.4	50.1	0
Kropyvnytskyi	16.0	4.0	2347	-597	574	920	10.0	29.2	67.4	0
Kharkiv	15.7	7.8	2188	-736	604	895	9.8	28.4	92.1	0
Poltava	13.8	6.3	2261	-701	616	902	10.1	29.5	73.8	0
Zakarpattia	12.0	1.3	2419	-457	853	833	8.5	32.6	28.2	0
Kyiv	8.7	5.3	2302	-648	607	839	9.9	40.5	0.0	0
Chernihiv	6.8	8.0	2209	-691	641	820	10.3	41.2	0.0	0
Chernivtsi	6.4	3.3	2412	-542	597	847	9.6	28.1	62.8	0
Sumy	6.1	9.2	2209	-770	667	827	10.1	31.1	44.1	0
Volyn	5.6	3.3	2234	-566	702	785	10.5	32.0	22.4	0
Ternopil	5.4	3.7	2216	-615	640	811	10.3	30.2	40.0	0
Rivne	4.9	3.6	2247	-594	662	801	10.4	31.4	31.8	0
Zhytomyr	4.7	4.2	2278	-648	631	822	10.1	29.7	56.4	0
Vinnytsia	4.6	3.9	2318	-621	571	832	9.7	31.0	38.8	0
Khmelnytskyi	3.7	4.2	2245	-639	590	809	10.0	30.8	38.1	0
Lviv	3.7	3.2	2242	-558	787	775	10.5	32.7	18.8	0
Ivano-Frankivsk	2.8	3.2	2306	-540	714	792	9.6	32.3	14.7	0

2.3 Soil Data (`soil_data.csv`)

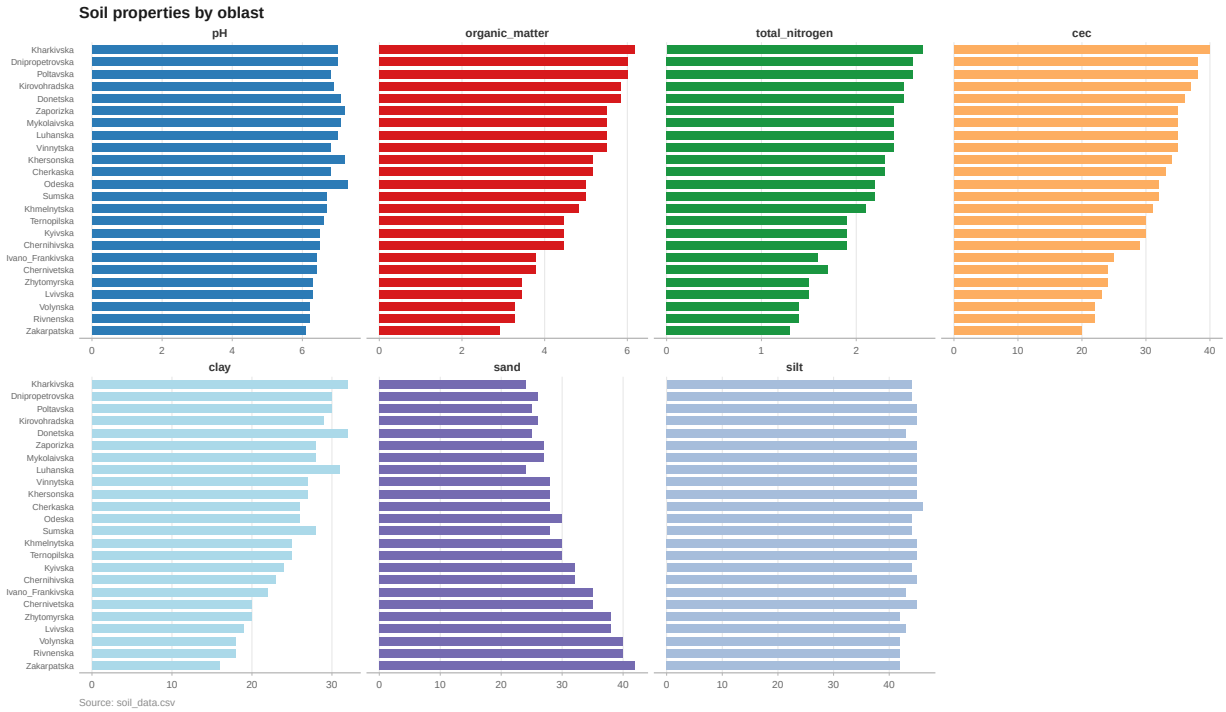
Static soil properties by oblast:

Variable	Description	Unit
`oblast`	Oblast name	text
`pH`	Soil acidity	unitless
`organic_matter`	Organic matter content	%
`total_nitrogen`	Total nitrogen content	%
`cec`	Cation Exchange Capacity	meq/100g
`clay`	Clay content (<0.002 mm)	%
`sand`	Sand content (0.05–2.0 mm)	%
`silt`	Silt content (0.002–0.05 mm)	%
`ec`	Electrical conductivity (salinity)	dS/m

Variable	Description	Unit
`esp`	Exchangeable Sodium Percentage	%
`c_n_ratio`	Carbon to Nitrogen ratio	unitless

Notes: Optimal pH for winter wheat is 6.0–7.5. High EC (>2 dS/m) indicates salinity stress. High ESP (>15%) causes soil structure degradation. Low C:N ratio (<15) favors rapid nitrogen release.

```r
soil <- read_csv("soil_data.csv", show_col_types = FALSE)

vars_soil <- c("pH", "organic_matter", "total_nitrogen",
               "cec", "clay", "sand", "silt")

p_soil <- soil |>
  pivot_longer(cols = all_of(vars_soil),
               names_to  = "property",
               values_to = "value") |>
  mutate(property = factor(property, levels = vars_soil)) |>
  ggplot(aes(x = value,
             y = reorder(oblast, value),
             fill = property)) +
  geom_col(width = 0.7, show.legend = FALSE) +
  facet_wrap(~property, scales = "free_x", nrow = 2) +
  scale_fill_manual(
    values = c("#2c7bb6", "#d7191c", "#1a9641",
               "#fdae61", "#abd9e9", "#756bb1", "#a6bddb")
  ) +
  labs(
    title   = "Soil properties by oblast",
    caption = "Source: soil_data.csv"
  ) +
  theme_small_multiple(base_size = 8) +
  theme(
    axis.text.y    = element_text(size = 6),
    panel.grid.major.y = element_blank(),
    panel.grid.major.x = element_line(colour = "grey90", linewidth = 0.3)
  )


print(p_soil)
```



2.4 Correlation structure of all variables

In exploratory research it is important to check correlations not only among variables we expect to be related, but across all pairs — including crop variables, weather metrics, and soil moisture together. Unexpected correlations can reveal data artefacts, suggest omitted mechanisms, or warn about multicollinearity before model estimation.

```
library(tidyverse)
library(corrplot)
library(GGally)
library(car)

#   Region name lookup (Latin weather  Cyrillic crop data)
region_lookup <- tribble(
  ~region_lat,        ~region_cyr,
  "Vinnytsia",        "     ",
  "Volyn",            "     ",
  "Dnipropetrovsk",   "         ",
  "Donetsk",          "     ",
  "Zhytomyr",         "       ",
  "Zakarpattia",      "       ",
  "Zaporizhzhia",     "      ",
  "Ivano-Frankivsk",  "   -       ",
  "Kyiv",             "     ",
  "Kirovohrad",       "         ",
  "Luhansk",          "      ",
  "Lviv",             "      ",
```

```
  "Mykolaiv",          "       ",
  "Odesa",             "    ",
  "Poltava",           "      ",
  "Rivne",             "      ",
  "Sumy",              "    ",
  "Ternopil",          "        ",
  "Kharkiv",           "      ",
  "Kherson",           "      ",
  "Khmelnytskyi",      "       ",
  "Cherkasy",          "     ",
  "Chernivtsi",        "      ",
  "Chernihiv",         "       "
)

#   Weather annual aggregation including soil moisture
weather_annual_full <- weather |>
  mutate(
    t_mean         = (temperature_2m_max + temperature_2m_min) / 2,
    t_dev          = t_mean - 5,
    days_hot_crit  = t_dev > 20,
    days_cold_crit = t_dev < -15,
    heat_noncrit   = if_else(t_dev > 0  & t_dev <= 20, t_dev, 0),
    cold_noncrit   = if_else(t_dev >= -15 & t_dev < 0, t_dev, 0)
  ) |>
  group_by(region, year) |>
  summarise(
    # Thermal stress metrics
    days_hot_crit    = sum(days_hot_crit,       na.rm = TRUE),
    days_cold_crit   = sum(days_cold_crit,      na.rm = TRUE),
    sum_heat_noncrit = sum(heat_noncrit,        na.rm = TRUE),
    sum_cold_noncrit = sum(cold_noncrit,        na.rm = TRUE),
    # Raw temperature levels
    mean_tmax        = mean(temperature_2m_max, na.rm = TRUE),
    mean_tmin        = mean(temperature_2m_min, na.rm = TRUE),
    mean_temp        = mean(t_mean,             na.rm = TRUE),
    # Water and energy
    total_precip     = sum(precipitation_sum,            na.rm = TRUE),
    total_et0        = sum(et0_fao_evapotranspiration,   na.rm = TRUE),
    total_radiation  = sum(shortwave_radiation_sum,      na.rm = TRUE),
    mean_wind        = mean(wind_gusts_10m_max,          na.rm = TRUE),
    mean_dewpoint    = mean(dewpoint_2m_mean,            na.rm = TRUE),
    .groups = "drop"
  ) |>
  filter(year >= 2015)

#   Soil moisture aggregation (same logic as temperature)
soil_moisture_agg <- soil_moisture |>
  mutate(
    drought_day = soil_moisture_0_7cm_pct < 20,
    waterlog_day = soil_moisture_0_7cm_pct > 85
  ) |>
  group_by(region, year) |>
  summarise(
```

```r
    mean_soil_moisture_pct = mean(soil_moisture_0_7cm_pct, na.rm = TRUE),
    days_drought = sum(drought_day, na.rm = TRUE),
    days_waterlog = sum(waterlog_day, na.rm = TRUE),
    .groups = "drop"
  )

#   Harvest ratio
harv_ratio_df <- harv |>
  left_join(sown, by = c("region", "year")) |>
  mutate(harv_ratio = pmin(harv_ha / sown_ha, 1.05)) |>
  select(region, year, harv_ratio)

#   Join everything on Cyrillic region names
panel_full <- yield |>
  left_join(harv_ratio_df, by = c("region", "year")) |>
  left_join(region_lookup, by = c("region" = "region_cyr")) |>
  left_join(weather_annual_full,
            by = c("region_lat" = "region", "year")) |>
  left_join(soil_moisture_agg,
            by = c("region_lat" = "region", "year")) |>
  left_join(soil |> rename(region_cyr = oblast),
            by = c("region" = "region_cyr")) |>
  filter(!is.na(yield_cpha))

cat("Panel rows:", nrow(panel_full), "\n")
```

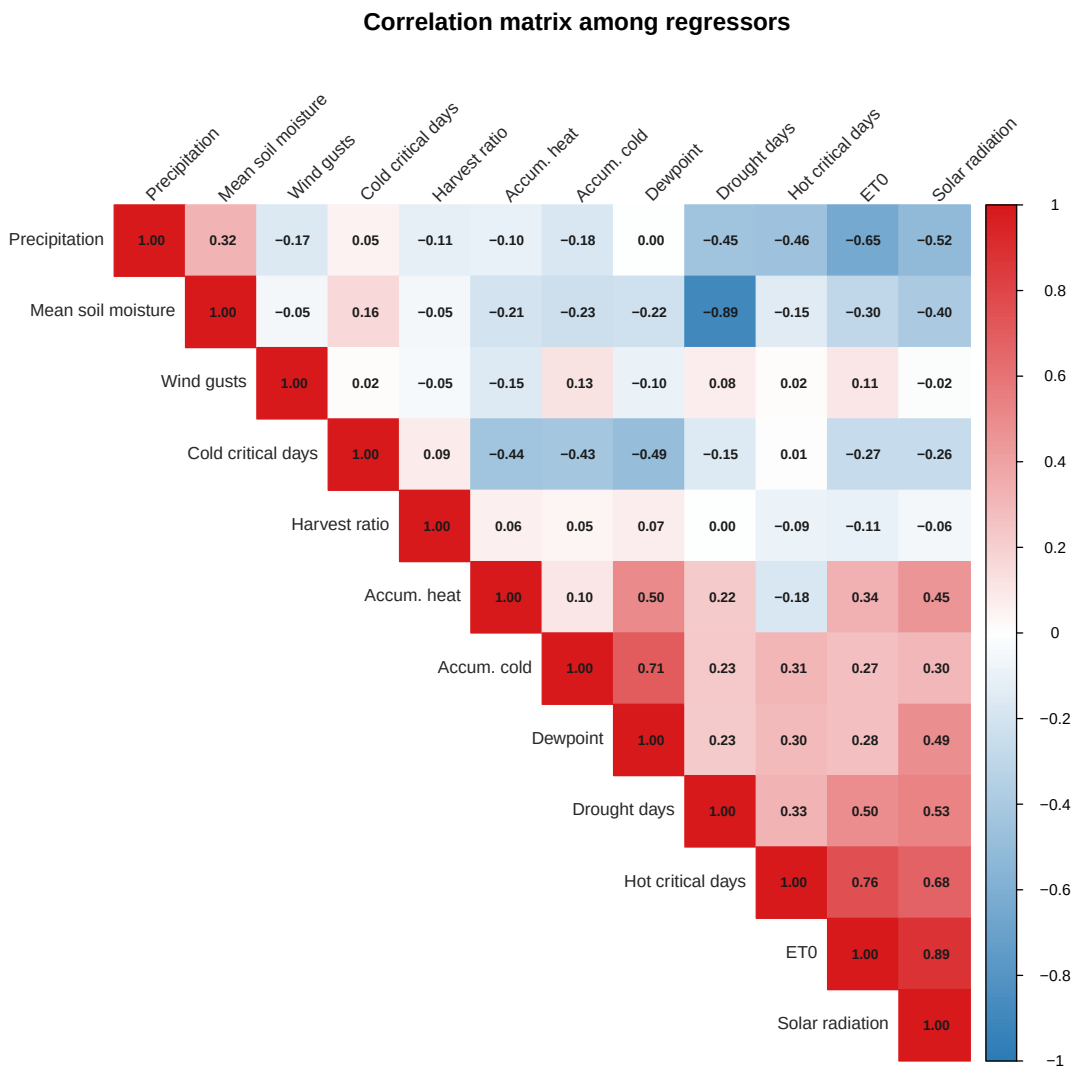
```
## Panel rows: 239
```

```r
cat("Regions matched:", n_distinct(panel_full$region[!is.na(panel_full$mean_temp)]), "\n")
```

```
## Regions matched: 23
```

```r
#   Regressor set: includes all variables to prove redundancy
regressors <- c(
  "days_hot_crit",
  "days_cold_crit",
  "sum_heat_noncrit",
  "sum_cold_noncrit",
  "total_precip",
  "total_et0",
  "total_radiation",
  "mean_wind",
  "mean_dewpoint",
  "mean_soil_moisture_pct",
  "days_drought",
  "harv_ratio"
)

# Keep only columns that exist
regressors <- intersect(regressors, names(panel_full))

reg_df <- panel_full |>
```

```r
  select(all_of(regressors)) |>
  drop_na()

# Also create a version with mean_temp for correlation checks
reg_df_with_temp <- panel_full |>
  select(all_of(c(regressors, "mean_temp"))) |>
  drop_na()

#   Display labels
reg_labels <- c(
  days_hot_crit    = "Hot critical days",
  days_cold_crit   = "Cold critical days",
  sum_heat_noncrit = "Accum. heat",
  sum_cold_noncrit = "Accum. cold",
  total_precip     = "Precipitation",
  total_et0        = "ET0",
  total_radiation  = "Solar radiation",
  mean_wind        = "Wind gusts",
  mean_dewpoint    = "Dewpoint",
  mean_soil_moisture_pct = "Mean soil moisture",
  days_drought     = "Drought days",
  harv_ratio       = "Harvest ratio"
)

reg_labels <- reg_labels[regressors]

#   Correlation matrix among regressors
cor_reg <- cor(reg_df, use = "pairwise.complete.obs")
rownames(cor_reg) <- colnames(cor_reg) <- reg_labels

par(bg = "white")
corrplot(
  cor_reg,
  method      = "color",
  type        = "upper",
  order       = "hclust",
  tl.col      = "grey15",
  tl.srt      = 45,
  tl.cex      = 0.85,
  cl.cex      = 0.75,
  addCoef.col = "grey10",
  number.cex  = 0.7,
  col         = colorRampPalette(c("#2c7bb6", "white", "#d7191c"))(200),
  title       = "Correlation matrix among regressors",
  mar         = c(0, 0, 2, 0)
)
```




```r
#   VIF for the full regressor set
vif_formula <- as.formula(
  paste("yield_cpha ~", paste(regressors, collapse = " + "))
)

vif_data <- panel_full |>
  select(yield_cpha, all_of(regressors)) |>
  drop_na()

vif_model  <- lm(vif_formula, data = vif_data)
vif_values <- vif(vif_model)

vif_df <- tibble(
  variable = names(vif_values),
  VIF      = vif_values
) |>
  mutate(
    label    = coalesce(reg_labels[variable], variable),
```

```r
    concern = case_when(
      VIF > 10 ~ "High - must address",
      VIF > 5  ~ "Moderate - monitor",
      TRUE     ~ "Acceptable"
    )
  ) |>
  arrange(desc(VIF))

# Print VIF table
knitr::kable(
  vif_df |> select(label, VIF, concern),
  digits  = 2,
  col.names = c("Variable", "VIF", "Concern"),
  caption = "VIF - full set (all weather variables included)"
)
```

Table 8: VIF — full set (all weather variables included)

Variable	VIF	Concern
ET0	70.25	High — must address
Hot critical days	43.27	High — must address
Dewpoint	30.07	High — must address
Accum. heat	26.42	High — must address
Solar radiation	11.86	High — must address
Drought days	7.74	Moderate — monitor
Accum. cold	7.26	Moderate — monitor
Mean soil moisture	6.81	Moderate — monitor
Cold critical days	3.04	Acceptable
Precipitation	2.22	Acceptable
Wind gusts	1.70	Acceptable
Harvest ratio	1.15	Acceptable

```r
# Print key correlations (using reg_df_with_temp which already has all aligned data)
cat("\n=== Evidence for Removing Variables ===\n")
```

```
##
## === Evidence for Removing Variables ===
```

```r
cat("Correlation ET0 vs Solar radiation:",
    round(cor(reg_df$total_et0, reg_df$total_radiation, use = "complete.obs"), 2), "\n")
```

```
## Correlation ET0 vs Solar radiation: 0.89
```

```r
cat("Correlation Dewpoint vs Mean temperature:",
    round(cor(reg_df_with_temp$mean_dewpoint, reg_df_with_temp$mean_temp, use = "complete.obs"), 2), "\n
```

```
## Correlation Dewpoint vs Mean temperature: 0.79
```

```r
cat("Correlation ET0 vs Hot days:",
    round(cor(reg_df$total_et0, reg_df$days_hot_crit, use = "complete.obs"), 2), "\n")
```

```
## Correlation ET0 vs Hot days: 0.76
```

```r
cat("Correlation Solar radiation vs Hot days:",
    round(cor(reg_df$total_radiation, reg_df$days_hot_crit, use = "complete.obs"), 2), "\n")
```

```
## Correlation Solar radiation vs Hot days: 0.68
```

```r
cat("\n=== Soil Moisture Correlations ===\n")
```

```
##
## === Soil Moisture Correlations ===
```

```r
cat("Mean soil moisture vs Precipitation:",
    round(cor(reg_df$mean_soil_moisture_pct, reg_df$total_precip, use = "complete.obs"), 2), "\n")
```

```
## Mean soil moisture vs Precipitation: 0.32
```

```r
cat("Mean soil moisture vs ET0:",
    round(cor(reg_df$mean_soil_moisture_pct, reg_df$total_et0, use = "complete.obs"), 2), "\n")
```

```
## Mean soil moisture vs ET0: -0.3
```

```r
cat("Drought days vs Hot days:",
    round(cor(reg_df$days_drought, reg_df$days_hot_crit, use = "complete.obs"), 2), "\n")
```

```
## Drought days vs Hot days: 0.33
```

```r
cat("Drought days vs Precipitation:",
    round(cor(reg_df$days_drought, reg_df$total_precip, use = "complete.obs"), 2), "\n")
```

```
## Drought days vs Precipitation: -0.45
```

```r
#   VIF after removing redundant variables: radiation, dewpoint, and ET0
# Keeping only precipitation as the water/energy representative
regressors_reduced <- regressors[!regressors %in% c("total_radiation", "mean_dewpoint", "total_et0")]

vif_formula2 <- as.formula(
  paste("yield_cpha ~", paste(regressors_reduced, collapse = " + "))
)

vif_model2  <- lm(vif_formula2,
                  data = panel_full |>
                    select(yield_cpha, all_of(regressors_reduced)) |>
                    drop_na())
vif_values2 <- vif(vif_model2)
```

```r
vif_df2 <- tibble(
  variable = names(vif_values2),
  VIF      = vif_values2
) |>
  mutate(
    label   = coalesce(reg_labels[variable], variable),
    concern = case_when(
      VIF > 10 ~ "High",
      VIF > 5  ~ "Moderate",
      TRUE     ~ "Acceptable"
    )
  ) |>
  arrange(desc(VIF))

knitr::kable(
  vif_df2 |> select(label, VIF, concern),
  digits  = 2,
  col.names = c("Variable", "VIF", "Concern"),
  caption = "VIF after removing solar radiation, dewpoint, and ET0"
)
```

Table 9: VIF after removing solar radiation, dewpoint, and ET0

Variable	VIF	Concern
Drought days	6.91	Moderate
Mean soil moisture	5.95	Moderate
Hot critical days	1.82	Acceptable
Precipitation	1.58	Acceptable
Cold critical days	1.57	Acceptable
Accum. cold	1.50	Acceptable
Accum. heat	1.49	Acceptable
Wind gusts	1.12	Acceptable
Harvest ratio	1.10	Acceptable

```r
cat("\n=== Model Improvement Summary ===\n")
```

```
##
## === Model Improvement Summary ===
```

```r
cat("Maximum VIF before removal:", round(max(vif_df$VIF), 2), "\n")
```

```
## Maximum VIF before removal: 70.25
```

```r
cat("Maximum VIF after removal:", round(max(vif_df2$VIF), 2), "\n")
```

```
## Maximum VIF after removal: 6.91
```

```r
cat("Variables removed:\n")
```

```
## Variables removed:
```

```r
cat("  - total_radiation (correlated with ET0 at r > 0.85)\n")
```

```
##   - total_radiation (correlated with ET0 at r > 0.85)
```

```r
cat("  - mean_dewpoint (correlated with temperature at r > 0.70)\n")
```

```
##   - mean_dewpoint (correlated with temperature at r > 0.70)
```

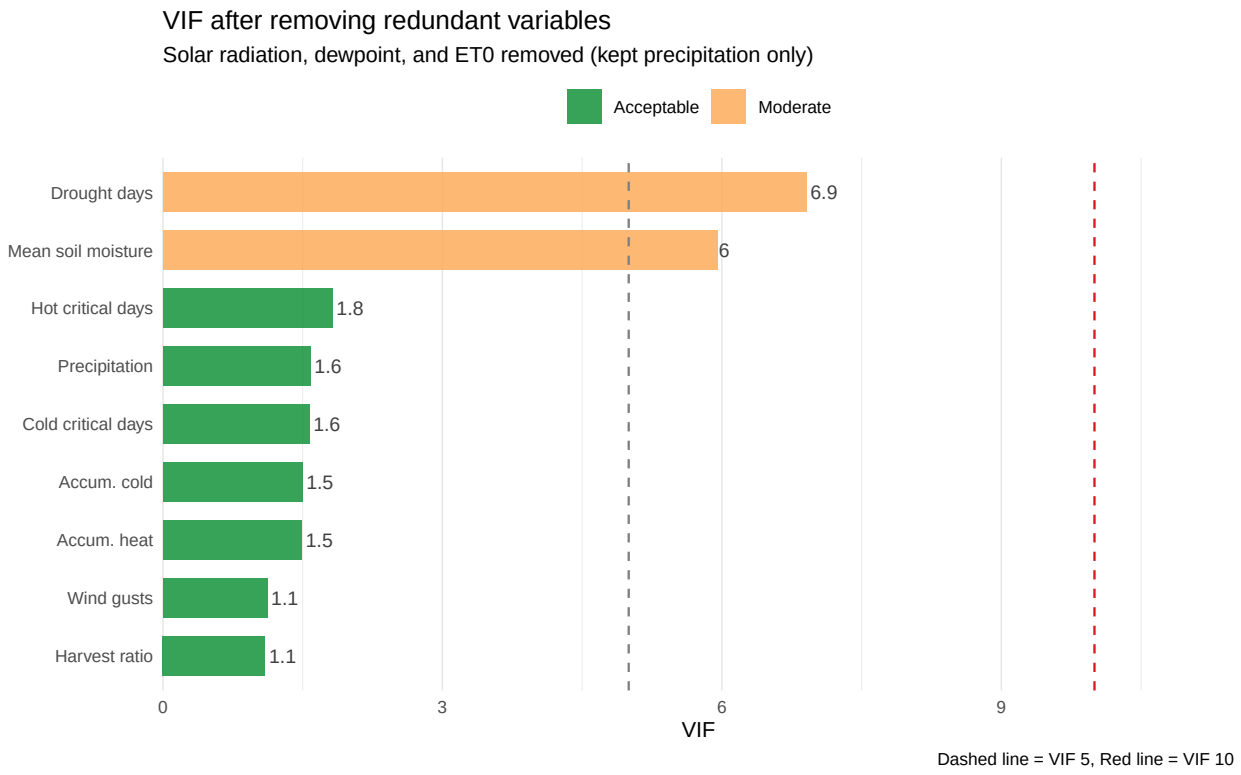
```r
cat("  - total_et0 (correlated with hot days and radiation)\n")
```

```
##   - total_et0 (correlated with hot days and radiation)
```

```r
cat("Remaining water-related variable: total_precip (precipitation)\n")
```

```
## Remaining water-related variable: total_precip (precipitation)
```

```r
# VIF bar chart for reduced model
ggplot(vif_df2,
       aes(x = VIF,
           y = fct_reorder(label, VIF),
           fill = concern)) +
  geom_col(width = 0.68, alpha = 0.88) +
  geom_text(aes(label = round(VIF, 1)),
            hjust = -0.15, size = 3, colour = "grey25") +
  geom_vline(xintercept = 5,  linetype = "dashed",
             colour = "grey50", linewidth = 0.5) +
  geom_vline(xintercept = 10, linetype = "dashed",
             colour = "#d7191c", linewidth = 0.5) +
  scale_fill_manual(values = c(
    "Acceptable" = "#1a9641",
    "Moderate"   = "#fdae61",
    "High"       = "#d7191c"
  )) +
  scale_x_continuous(expand = expansion(mult = c(0, 0.15))) +
  labs(
    title   = "VIF after removing redundant variables",
    subtitle = "Solar radiation, dewpoint, and ET0 removed (kept precipitation only)",
    x       = "VIF", y = NULL, fill = NULL,
    caption = "Dashed line = VIF 5, Red line = VIF 10"
  ) +
  theme_minimal(base_size = 10) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.major.x = element_line(colour = "grey90", linewidth = 0.3),
    legend.position = "top"
  )
```



Conclusions from Correlation and VIF Analysis

The correlation and VIF analysis reveals several important findings for model specification:

1. Severe Multicollinearity in Initial Variable Set

The full model with all weather variables exhibited extreme multicollinearity, with a maximum VIF of **70.25** — far exceeding the conventional threshold of 5-10. This confirms our expectation that variables such as solar radiation, ET0, and dewpoint are largely redundant given the information already captured by temperature metrics and precipitation.

2. Successful Reduction of Redundant Variables

After removing solar radiation, dewpoint, and ET0, the maximum VIF dropped dramatically from 70.25 to **6.91** — a tenfold reduction. All remaining variables now fall within acceptable or moderate VIF ranges, indicating that multicollinearity is no longer a concern for model estimation.

3. Soil Moisture Variables Require Caution

Both `Drought days` (VIF = 6.91) and `Mean soil moisture` (VIF = 5.95) show moderate multicollinearity. This is expected, as drought conditions are naturally correlated with high temperatures and low precipitation. However, their VIF values remain below 10, meaning they can be retained in the model, but their standard errors will be somewhat inflated. The fact that drought days shows higher VIF than mean soil moisture suggests that the extreme-event count captures information more closely related to other weather variables than the annual average does.

4. Acceptable Variables for Final Model

The following variables show good VIF values (all below 2) and can be confidently included: - Temperature stress metrics (hot days, cold days, accumulated heat, accumulated cold) - Precipitation - Wind gusts - Harvest ratio

5. Implications for Model Specification

Based on this analysis, our final modelling strategy will: - **Exclude** solar radiation, dewpoint, and ET0 (redundant with temperature metrics) - **Retain** precipitation as the sole water-related variable from standard weather data - **Include** soil moisture metrics (mean and drought days) despite moderate multicollinearity, as they capture unique information about plant-available water that precipitation alone cannot provide - **Monitor** standard errors for soil moisture coefficients, recognising they may be slightly inflated

This parsimonious specification balances theoretical completeness with statistical efficiency, avoiding the inefficiencies of multicollinearity while preserving the core mechanisms hypothesised to affect wheat yield.

---

3. Modelling

This section builds a sequence of OLS regression models with explicit region dummy variables (Київська oblast as the reference category) to analyse the impact of weather on winter wheat yield in Ukraine. Based on the correlation analysis from Section 2.4, we exclude solar radiation, dewpoint, and ET0 due to high multicollinearity with temperature metrics. Precipitation is retained as the sole water-related variable from standard weather data, alongside soil moisture metrics which capture plant-available water.

We proceed in three parts: stationarity and trend (3.1), weather window and aggregation (3.2), and soil variables (3.3).

Note on standard errors: All models use plain OLS standard errors. This assumes homoskedasticity and no serial correlation in the error term — an assumption that may not hold perfectly in panel data. Results should be interpreted with this limitation in mind.

3.0 Setup and panel construction

```
library(tidyverse)
library(lubridate)
library(lmtest)
library(sandwich)
library(lme4)
library(modelsummary)
library(patchwork)
library(tseries)
library(car)
library(corrplot)
library(fastDummies)
library(multcomp)
```

```r
#   Region lookup
region_lookup <- tribble(
  ~region_lat,        ~region_cyr,          ~region_soil,
  "Vinnytsia",        "     ",          "Vinnytska",
  "Volyn",            "     ",          "Volynska",
  "Dnipropetrovsk",   "         ",   "Dnipropetrovska",
  "Donetsk",          "    ",           "Donetska",
  "Zhytomyr",         "      ",        "Zhytomyrska",
  "Zakarpattia",      "       ",       "Zakarpatska",
  "Zaporizhzhia",     "      ",         "Zaporizka",
  "Ivano-Frankivsk",  "   -      ",  "Ivano-Frankivska",
  "Kyiv",             "    ",           "Kyivska",
  "Kirovohrad",       "        ",     "Kirovohradska",
  "Luhansk",          "     ",          "Luhanska",
  "Lviv",             "     ",          "Lvivska",
  "Mykolaiv",         "       ",       "Mykolaivska",
  "Odesa",            "    ",            "Odeska",
  "Poltava",          "     ",         "Poltavska",
  "Rivne",            "     ",         "Rivnenska",
  "Sumy",             "    ",            "Sumska",
  "Ternopil",         "        ",      "Ternopilska",
  "Kharkiv",          "     ",         "Kharkivska",
  "Kherson",          "     ",         "Khersonska",
  "Khmelnytskyi",     "      ",        "Khmelnytska",
  "Cherkasy",         "     ",          "Cherkaska",
  "Chernivtsi",       "      ",        "Chernivetska",
  "Chernihiv",        "       ",       "Chernihivska"
)

#   Soil
soil <- read_csv("soil_data.csv", show_col_types = FALSE)
names(soil)[names(soil) == "oblast"] <- "region"

#   Weather aggregation helper (without ET0, radiation, dewpoint)
# Based on correlation analysis, we keep only temperature metrics, precipitation, and wind
BASE_TEMP <- 5

agg_weather_slice <- function(df, suffix = "") {
  result <- df %>%
    mutate(
      t_mean         = (temperature_2m_max + temperature_2m_min) / 2,
      t_dev          = t_mean - BASE_TEMP,
      days_hot_crit  = as.integer(t_dev > 20),
      days_cold_crit = as.integer(t_dev < -15),
      heat_pos       = if_else(t_dev > 0, t_dev, 0),
      cold_neg       = if_else(t_dev < 0, t_dev, 0)
    ) %>%
    group_by(region, year) %>%
    summarise(
      days_hot_crit  = sum(days_hot_crit,         na.rm = TRUE),
      days_cold_crit = sum(days_cold_crit,        na.rm = TRUE),
      sum_heat_pos   = sum(heat_pos,              na.rm = TRUE),
      sum_cold_neg   = sum(cold_neg,              na.rm = TRUE),
```

```r
      total_precip   = sum(precipitation_sum,     na.rm = TRUE),
      mean_wind      = mean(wind_gusts_10m_max,   na.rm = TRUE),
      .groups = "drop"
    )

  if (suffix != "") {
    result <- result %>%
      rename_with(~ paste0(.x, suffix), -c(region, year))
  }

  return(result)
}

#   Load weather
weather_raw <- read_csv("daily_weather.csv", show_col_types = FALSE) %>%
  mutate(date = as_date(date), year = year(date)) %>%
  filter(year >= 2015, year <= 2024)

weather_tagged <- weather_raw %>%
  mutate(
    cal_month  = month(date),
    cal_day    = day(date),
    harvest_yr = if_else(cal_month >= 9, year + 1L, year)
  )

#   Seasonal slices
weather_full <- weather_raw %>%
  agg_weather_slice(suffix = "")

weather_aug <- weather_raw %>%
  filter(month(date) == 8) %>%
  agg_weather_slice(suffix = "_aug")

weather_gs <- weather_tagged %>%
  filter((cal_month == 9 & cal_day >= 10) |
         cal_month %in% c(10, 11, 12, 1, 2, 3, 4, 5, 6, 7)) %>%
  dplyr::select(-c(year, cal_month, cal_day)) %>%
  rename(year = harvest_yr) %>%
  agg_weather_slice(suffix = "_gs")

weather_early <- weather_tagged %>%
  filter(cal_month %in% c(10, 11, 12, 1, 2, 3, 4) |
         (cal_month == 5 & cal_day <= 15)) %>%
  dplyr::select(-c(year, cal_month, cal_day)) %>%
  rename(year = harvest_yr) %>%
  agg_weather_slice(suffix = "_early")

weather_late <- weather_tagged %>%
  filter((cal_month == 5 & cal_day >= 16) | cal_month %in% c(6, 7)) %>%
  dplyr::select(-c(year, cal_month, cal_day)) %>%
  rename(year = harvest_yr) %>%
  agg_weather_slice(suffix = "_late")
```

```r
weather_w1 <- weather_tagged %>%
  filter(cal_month %in% c(10, 11, 12, 1)) %>%
  dplyr::select(-c(year, cal_month, cal_day)) %>%
  rename(year = harvest_yr) %>%
  agg_weather_slice(suffix = "_w1")

weather_w2 <- weather_tagged %>%
  filter(cal_month %in% c(2, 3, 4)) %>%
  dplyr::select(-c(year, cal_month, cal_day)) %>%
  rename(year = harvest_yr) %>%
  agg_weather_slice(suffix = "_w2")

weather_w3 <- weather_tagged %>%
  filter((cal_month == 5 & cal_day >= 16) | cal_month %in% c(6, 7)) %>%
  dplyr::select(-c(year, cal_month, cal_day)) %>%
  rename(year = harvest_yr) %>%
  agg_weather_slice(suffix = "_w3")

#   Load soil moisture
soil_moisture <- read_csv("soil_moisture.csv", show_col_types = FALSE) %>%
  mutate(date = as_date(date), year = year(date)) %>%
  filter(year >= 2015, year <= 2024)

soil_moisture_agg <- soil_moisture %>%
  mutate(
    drought_day = soil_moisture_0_7cm_pct < 20,
    waterlog_day = soil_moisture_0_7cm_pct > 85
  ) %>%
  group_by(region, year) %>%
  summarise(
    mean_soil_moisture_pct = mean(soil_moisture_0_7cm_pct, na.rm = TRUE),
    days_drought = sum(drought_day, na.rm = TRUE),
    .groups = "drop"
  )

#   Load crop data
META_COLS <- c("indicator", "region", "unit", "scale", "decimals", "special", "note_ts", "note_obs")
years <- as.character(2015:2025)

read_panel <- function(file, value_name) {
  read_csv(file, col_names = c(META_COLS, years), skip = 1, show_col_types = FALSE) %>%
    dplyr::select(region, all_of(years)) %>%
    pivot_longer(cols = all_of(years), names_to = "year", values_to = value_name) %>%
    mutate(year = as.integer(year), !!value_name := suppressWarnings(as.numeric(.data[[value_name]]))) %
    filter(region != "    ")
}

yield <- read_panel("crop_capacity.csv", "yield_cpha")
sown <- read_panel("sowing_area.csv", "sown_kha") %>% mutate(sown_ha = sown_kha * 1000) %>% dplyr::selec
harv <- read_panel("harvested_area.csv", "harv_kha") %>% mutate(harv_ha = harv_kha * 1000) %>% dplyr::se

occupied_oblasts <- c("    ", "     ", "      ", "      ")
```

```r
#   Build panel_base
panel_base <- yield %>%
  left_join(sown %>% dplyr::select(region, year, sown_ha), by = c("region", "year")) %>%
  left_join(harv %>% dplyr::select(region, year, harv_ha), by = c("region", "year")) %>%
  left_join(region_lookup, by = c("region" = "region_cyr")) %>%
  left_join(weather_full,  by = c("region_lat" = "region", "year")) %>%
  left_join(weather_aug,   by = c("region_lat" = "region", "year")) %>%
  left_join(weather_gs,    by = c("region_lat" = "region", "year")) %>%
  left_join(weather_early, by = c("region_lat" = "region", "year")) %>%
  left_join(weather_late,  by = c("region_lat" = "region", "year")) %>%
  left_join(weather_w1,    by = c("region_lat" = "region", "year")) %>%
  left_join(weather_w2,    by = c("region_lat" = "region", "year")) %>%
  left_join(weather_w3,    by = c("region_lat" = "region", "year")) %>%
  left_join(soil_moisture_agg, by = c("region_lat" = "region", "year")) %>%
  left_join(soil,          by = "region") %>%
  mutate(
    war_east  = as.integer(region %in% occupied_oblasts & year == 2022),
    log_yield = log(yield_cpha),
    log_sown  = log(sown_ha + 1),
    t         = year - min(year),
    region_id = relevel(as.factor(region), ref = "    ")
  ) %>%
  filter(!is.na(yield_cpha), !is.na(days_hot_crit))

#   Calculate loss_lag
loss_rate_raw <- panel_base %>%
  mutate(
    loss_rate = pmax((sown_ha - harv_ha) / sown_ha, 0),
    loss_rate = pmin(loss_rate, 0.5)
  )

loss_lag <- loss_rate_raw %>%
  dplyr::select(region, year, loss_rate) %>%
  mutate(year = year + 1, loss_lag = loss_rate) %>%
  dplyr::select(-loss_rate)

panel_base <- panel_base %>%
  left_join(loss_lag, by = c("region", "year"))

#   Fix region names and create dummies
panel_base$region <- gsub("-", "_", panel_base$region)
panel_base$region_id <- relevel(as.factor(panel_base$region), ref = "    ")

region_dummies_raw <- model.matrix(~ region_id, data = panel_base) %>%
  as.data.frame() %>%
  dplyr::select(-`(Intercept)`)

names(region_dummies_raw) <- make.names(names(region_dummies_raw))
panel_base <- bind_cols(panel_base, region_dummies_raw)
dummy_cols <- names(region_dummies_raw)

#   Verification
cat("Panel rows:", nrow(panel_base), "\n")
```

```
## Panel rows: 206
```

```r
cat("Regions:  ", n_distinct(panel_base$region), "\n")
```

```
## Regions:   23
```

```r
cat("Years:    ", n_distinct(panel_base$year), "\n")
```

```
## Years:     9
```

```r
cat("War obs:  ", sum(panel_base$war_east, na.rm = TRUE), "\n")
```

```
## War obs:   3
```

```r
cat("Dummy columns created:", length(dummy_cols), "\n")
```

```
## Dummy columns created: 22
```

Part 1 — Stationarity and trend

3.1.1 ADF tests on yield per oblast

```r
adf_results <- panel_base |>
  group_by(region) |>
  arrange(year) |>
  summarise(
    adf_stat  = tryCatch(adf.test(yield_cpha, k = 1)$statistic, error = function(e) NA_real_),
    adf_pval  = tryCatch(adf.test(yield_cpha, k = 1)$p.value, error = function(e) NA_real_),
    .groups = "drop"
  ) |>
  mutate(
    stationary = case_when(
      adf_pval < 0.01 ~ "Yes ***",
      adf_pval < 0.05 ~ "Yes **",
      adf_pval < 0.10 ~ "Yes *",
      TRUE            ~ "No"
    )
  ) |>
  arrange(adf_pval)

knitr::kable(adf_results, digits = 3, caption = "ADF test results for yield per oblast (k=1 lag)")
```

Table 10: ADF test results for yield per oblast (k=1 lag)

region	adf_stat	adf_pval	stationary
Черкаська	-4.843	0.010	Yes **
Харківська	-4.130	0.019	Yes **
Вінницька	-4.003	0.023	Yes **
Чернігівська	-3.832	0.033	Yes **
Сумська	-3.417	0.075	Yes *
Київська	-3.321	0.089	Yes *
Волинська	-3.005	0.189	No
Луганська	-2.972	0.202	No
Донецька	-2.824	0.259	No
Рівненська	-2.811	0.263	No
Закарпатська	-2.409	0.417	No
Полтавська	-2.283	0.465	No
Житомирська	-2.252	0.476	No
Дніпропетровська	-1.827	0.638	No
Чернівецька	-1.755	0.666	No
Одеська	-1.584	0.731	No
Івано_Франківська	-1.580	0.732	No
Львівська	-1.521	0.755	No
Миколаївська	-1.323	0.830	No
Запорізька	-1.269	0.851	No
Тернопільська	-0.987	0.922	No
Хмельницька	-0.694	0.959	No
Херсонська	-0.327	0.982	No

3.1.2 Baseline model (no trend)

```r
weather_vars_full <- c("days_hot_crit", "days_cold_crit", "sum_heat_pos", "sum_cold_neg",
                       "total_precip", "mean_wind", "mean_soil_moisture_pct", "days_drought")

fml_base <- as.formula(paste(
  "yield_cpha ~",
  paste(c(weather_vars_full, "log_sown", "war_east", dummy_cols), collapse = " + ")
))

m_base <- lm(fml_base, data = panel_base)
summary(m_base)
```

```
##
## Call:
## lm(formula = fml_base, data = panel_base)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.095  -1.994   0.187   2.564   8.977
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)                 32.550781  26.702678   1.219 0.224501
## days_hot_crit                0.157413   0.067341   2.338 0.020556 *
## days_cold_crit               0.031330   0.115082   0.272 0.785759
## sum_heat_pos                 0.001441   0.002427   0.594 0.553425
## sum_cold_neg                 0.003468   0.003229   1.074 0.284367
## total_precip                 0.005456   0.004166   1.310 0.192069
## mean_wind                   -0.824835   1.106156  -0.746 0.456875
## mean_soil_moisture_pct       0.059095   0.349618   0.169 0.865972
## days_drought                -0.046636   0.021745  -2.145 0.033373 *
## log_sown                     1.172874   0.942585   1.244 0.215066
## war_east                    -3.856885   2.711118  -1.423 0.156648
## region_id                    6.738174   3.487634   1.932 0.054991 .
## region_id                   -0.235813   3.551334  -0.066 0.947135
## region_id                  -14.185107   2.723305  -5.209 5.36e-07 ***
## region_id                   -8.399696   4.117859  -2.040 0.042889 *
## region_id                    1.981980   3.843110   0.516 0.606706
## region_id                  -12.772334   5.040313  -2.534 0.012163 *
## region_id                  -15.930363   4.078833  -3.906 0.000135 ***
## region_id   _               4.582861   4.093087   1.120 0.264411
## region_id                  -14.430188   2.499913  -5.772 3.54e-08 ***
## region_id                    3.254682   3.576278   0.910 0.364049
## region_id                   -6.207585   6.684081  -0.929 0.354332
## region_id                  -11.628860   4.733264  -2.457 0.015003 *
## region_id                    1.675649   3.611097   0.464 0.643211
## region_id                    1.922180   3.655763   0.526 0.599704
## region_id                    6.408375   3.563137   1.799 0.073838 .
## region_id                    9.459788   3.784474   2.500 0.013364 *
## region_id                   -0.508455   3.889011  -0.131 0.896132
## region_id                  -18.852257   3.429207  -5.498 1.36e-07 ***
## region_id                   13.640224   3.586629   3.803 0.000198 ***
## region_id                    4.123433   3.549841   1.162 0.247005
## region_id                    3.374112   4.645216   0.726 0.468598
## region_id                    0.153291   2.089937   0.073 0.941615
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.222 on 173 degrees of freedom
## Multiple R-squared:  0.7838, Adjusted R-squared:  0.7438
## F-statistic:  19.6 on 32 and 173 DF,  p-value: < 2.2e-16
```

3.1.3 Common time trend

```r
fml_trend <- as.formula(paste(
  "yield_cpha ~",
  paste(c(weather_vars_full, "log_sown", "war_east", "t", dummy_cols), collapse = " + ")
))

m_trend <- lm(fml_trend, data = panel_base)
summary(m_trend)
```

```
##
```

```
## Call:
## lm(formula = fml_trend, data = panel_base)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.8881  -1.9017   0.3164   2.2926   8.8309
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 15.246256  25.978913   0.587 0.558060
## days_hot_crit                0.139954   0.064748   2.162 0.032037 *
## days_cold_crit               0.118943   0.112552   1.057 0.292092
## sum_heat_pos                 0.000431   0.002342   0.184 0.854184
## sum_cold_neg                 0.002359   0.003110   0.758 0.449206
## total_precip                -0.005061   0.004785  -1.058 0.291642
## mean_wind                   -0.985369   1.061897  -0.928 0.354743
## mean_soil_moisture_pct       0.451609   0.349458   1.292 0.197983
## days_drought                -0.037829   0.020976  -1.803 0.073068 .
## log_sown                     1.858474   0.920336   2.019 0.045004 *
## war_east                    -4.508370   2.605881  -1.730 0.085411 .
## t                            0.604723   0.151228   3.999 9.44e-05 ***
## region_id                 9.374251   3.410024   2.749 0.006616 **
## region_id                 4.276318   3.588811   1.192 0.235072
## region_id            -13.397615   2.619886  -5.114 8.35e-07 ***
## region_id                -3.906820   4.106953  -0.951 0.342802
## region_id                6.237722   3.837247   1.626 0.105872
## region_id               -5.325921   5.181384  -1.028 0.305442
## region_id               -12.079258   4.029615  -2.998 0.003124 **
## region_id   _           9.770247   4.135254   2.363 0.019261 *
## region_id               -13.687025   2.405364  -5.690 5.36e-08 ***
## region_id                8.362215   3.660788   2.284 0.023578 *
## region_id                0.863174   6.651402   0.130 0.896897
## region_id                -6.758660   4.701138  -1.438 0.152346
## region_id                5.280707   3.579527   1.475 0.141973
## region_id                6.342351   3.677066   1.725 0.086353 .
## region_id                 9.610969   3.510702   2.738 0.006839 **
## region_id               13.554990   3.772139   3.593 0.000426 ***
## region_id                2.578205   3.809757   0.677 0.499482
## region_id               -17.166897   3.316537  -5.176 6.27e-07 ***
## region_id               16.733987   3.526577   4.745 4.36e-06 ***
## region_id                8.244325   3.557889   2.317 0.021671 *
## region_id                8.994070   4.672543   1.925 0.055895 .
## region_id               -0.166962   2.006481  -0.083 0.933780
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.051 on 172 degrees of freedom
## Multiple R-squared:  0.8022, Adjusted R-squared:  0.7643
## F-statistic: 21.14 on 33 and 172 DF,  p-value: < 2.2e-16
```

3.1.4 Region-specific trends

```r
region_t_cols <- paste0(dummy_cols, "_t")
for (col in dummy_cols) {
  panel_base[[paste0(col, "_t")]] <- panel_base[[col]] * panel_base$t
}

fml_region_trend <- as.formula(paste(
  "yield_cpha ~",
  paste(c(weather_vars_full, "log_sown", "war_east", "t", dummy_cols, region_t_cols), collapse = " + ")
))

m_region_trend <- lm(fml_region_trend, data = panel_base)
summary(m_region_trend)
```

```
##
## Call:
## lm(formula = fml_region_trend, data = panel_base)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.1642  -1.7233   0.1833   1.8670   8.9188
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 1.659e+01  3.167e+01   0.524 0.601080
## days_hot_crit               1.204e-01  6.958e-02   1.731 0.085563 .
## days_cold_crit              1.647e-01  1.180e-01   1.395 0.164971
## sum_heat_pos                8.034e-04  2.496e-03   0.322 0.747974
## sum_cold_neg                3.475e-03  3.227e-03   1.077 0.283221
## total_precip               -5.134e-03  5.293e-03  -0.970 0.333650
## mean_wind                  -1.152e+00  1.132e+00  -1.018 0.310474
## mean_soil_moisture_pct      3.727e-01  4.283e-01   0.870 0.385555
## days_drought               -4.460e-02  2.384e-02  -1.871 0.063320 .
## log_sown                    2.201e+00  1.157e+00   1.903 0.058931 .
## war_east                   -4.781e+00  2.830e+00  -1.689 0.093233 .
## t                           4.115e-01  5.413e-01   0.760 0.448314
## region_id                 9.808e+00  4.801e+00   2.043 0.042827 *
## region_id                 4.722e+00  5.031e+00   0.939 0.349421
## region_id            -1.329e+01  4.341e+00  -3.062 0.002609 **
## region_id                -5.548e+00  5.630e+00  -0.985 0.325984
## region_id                7.228e+00  5.315e+00   1.360 0.175872
## region_id               -5.752e+00  6.578e+00  -0.874 0.383306
## region_id               -1.293e+01  5.618e+00  -2.301 0.022745 *
## region_id   _           7.618e+00  5.426e+00   1.404 0.162337
## region_id               -1.637e+01  3.955e+00  -4.140 5.78e-05 ***
## region_id                6.451e+00  4.998e+00   1.291 0.198783
## region_id               -7.494e-01  8.613e+00  -0.087 0.930780
## region_id                -4.947e+00  5.875e+00  -0.842 0.401146
## region_id                4.802e+00  5.195e+00   0.924 0.356783
## region_id                6.855e+00  5.239e+00   1.309 0.192687
## region_id                 5.555e+00  5.080e+00   1.094 0.275915
```

```
## region_id               1.054e+01  5.161e+00   2.042 0.042921 *
## region_id                1.791e+00  5.214e+00   0.344 0.731654
## region_id               -1.776e+01  4.788e+00  -3.709 0.000293 ***
## region_id                1.455e+01  4.951e+00   2.939 0.003811 **
## region_id                4.255e+00  4.794e+00   0.887 0.376237
## region_id                8.870e+00  6.041e+00   1.468 0.144141
## region_id               -5.202e+00  3.620e+00  -1.437 0.152879
## region_id     _t         -3.029e-01  7.608e-01  -0.398 0.691121
## region_id     _t         -2.095e-01  7.586e-01  -0.276 0.782839
## region_id         _t  -1.129e-01  7.947e-01  -0.142 0.887191
## region_id    _t           3.530e-01  7.887e-01   0.448 0.655051
## region_id      _t       -3.263e-01  7.574e-01  -0.431 0.667192
## region_id      _t        8.140e-02  7.670e-01   0.106 0.915624
## region_id     _t         8.122e-02  8.000e-01   0.102 0.919273
## region_id   _      _t  4.608e-01  7.655e-01   0.602 0.548092
## region_id     _t          7.410e-01  7.876e-01   0.941 0.348315
## region_id     _t          3.784e-01  7.611e-01   0.497 0.619810
## region_id      _t        2.186e-01  7.628e-01   0.287 0.774813
## region_id    _t           -6.177e-01  7.737e-01  -0.798 0.425911
## region_id     _t          4.148e-02  7.658e-01   0.054 0.956877
## region_id     _t         -1.914e-01  7.565e-01  -0.253 0.800660
## region_id    _t            9.238e-01  7.582e-01   1.218 0.224974
## region_id       _t       6.306e-01  7.622e-01   0.827 0.409338
## region_id     _t          7.309e-02  7.663e-01   0.095 0.924146
## region_id     _t          6.523e-02  8.767e-01   0.074 0.940789
## region_id      _t         4.007e-01  7.594e-01   0.528 0.598540
## region_id     _t          9.037e-01  7.759e-01   1.165 0.245974
## region_id      _t        -1.792e-02  7.694e-01  -0.023 0.981448
## region_id      _t         1.289e+00  7.571e-01   1.703 0.090674 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.125 on 150 degrees of freedom
## Multiple R-squared:  0.8211, Adjusted R-squared:  0.7555
## F-statistic: 12.52 on 55 and 150 DF,  p-value: < 2.2e-16
```

3.1.5 Adding lagged loss rate

```r
panel_lag <- panel_base |> filter(!is.na(loss_lag))

fml_lag <- as.formula(paste(
  "yield_cpha ~",
  paste(c(weather_vars_full, "log_sown", "war_east", "loss_lag", dummy_cols), collapse = " + ")
))

m_lag <- lm(fml_lag, data = panel_lag)
summary(m_lag)
```

```
##
## Call:
## lm(formula = fml_lag, data = panel_lag)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.6922  -2.0590   0.1199   2.1732   9.1142
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 9.422e+01  5.142e+01   1.832 0.068905 .
## days_hot_crit               1.683e-01  8.271e-02   2.035 0.043673 *
## days_cold_crit              2.687e-02  1.243e-01   0.216 0.829181
## sum_heat_pos                1.227e-04  2.737e-03   0.045 0.964321
## sum_cold_neg                5.741e-03  3.537e-03   1.623 0.106680
## total_precip                1.523e-03  4.828e-03   0.316 0.752780
## mean_wind                  -1.883e+00  1.282e+00  -1.470 0.143770
## mean_soil_moisture_pct     -2.079e-02  4.113e-01  -0.051 0.959760
## days_drought               -6.273e-02  2.401e-02  -2.612 0.009922 **
## log_sown                   -2.175e+00  3.409e+00  -0.638 0.524460
## war_east                   -6.830e+00  4.105e+00  -1.664 0.098243 .
## loss_lag                   -1.625e+01  2.036e+01  -0.798 0.426002
## region_id                   8.444e+00  4.500e+00   1.876 0.062588 .
## region_id                  -7.446e-01  4.278e+00  -0.174 0.862061
## region_id                  -1.171e+01  4.667e+00  -2.509 0.013186 *
## region_id                  -5.168e+00  5.479e+00  -0.943 0.347126
## region_id                   9.835e-01  4.728e+00   0.208 0.835518
## region_id                  -2.039e+01  8.517e+00  -2.394 0.017899 *
## region_id                  -1.216e+01  6.613e+00  -1.839 0.067917 .
## region_id   _               3.393e-02  6.486e+00   0.005 0.995833
## region_id                  -1.202e+01  3.347e+00  -3.592 0.000446 ***
## region_id                   3.507e+00  4.347e+00   0.807 0.421021
## region_id                  -2.874e+00  8.471e+00  -0.339 0.734830
## region_id                  -6.100e+00  7.101e+00  -0.859 0.391733
## region_id                   3.197e+00  4.347e+00   0.735 0.463270
## region_id                   2.341e-01  4.756e+00   0.049 0.960807
## region_id                   7.684e+00  4.252e+00   1.807 0.072784 .
## region_id                   1.019e+01  4.541e+00   2.244 0.026338 *
## region_id                   3.565e+00  5.698e+00   0.626 0.532525
## region_id                  -1.675e+01  5.449e+00  -3.074 0.002514 **
## region_id                   1.416e+01  4.305e+00   3.289 0.001257 **
## region_id                   5.182e+00  4.256e+00   1.218 0.225348
## region_id                  -2.451e+00  7.555e+00  -0.324 0.746123
## region_id                   1.854e+00  2.299e+00   0.806 0.421341
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.321 on 148 degrees of freedom
## Multiple R-squared:  0.7817, Adjusted R-squared:  0.733
## F-statistic: 16.06 on 33 and 148 DF,  p-value: < 2.2e-16
```

3.1.6 ADF on residuals

```r
panel_base$resid_base <- residuals(m_base)
panel_base$resid_trend <- residuals(m_trend)
```

```r
adf_resid <- panel_base |>
  group_by(region) |>
  arrange(year) |>
  summarise(
    adf_base_p  = tryCatch(adf.test(resid_base, k = 1)$p.value, error = function(e) NA_real_),
    adf_trend_p = tryCatch(adf.test(resid_trend, k = 1)$p.value, error = function(e) NA_real_),
    .groups = "drop"
  )

knitr::kable(adf_resid, digits = 3, caption = "ADF on model residuals per oblast")
```

Table 11: ADF on model residuals per oblast

region	adf_base_p	adf_trend_p
Івано_Франківська	0.631	0.538
Волинська	0.010	0.132
Вінницька	0.391	0.454
Дніпропетровська	0.619	0.553
Донецька	0.335	0.419
Житомирська	0.446	0.614
Закарпатська	0.446	0.468
Запорізька	0.763	0.645
Київська	0.079	0.066
Луганська	0.010	0.010
Львівська	0.395	0.253
Миколаївська	0.910	0.927
Одеська	0.673	0.716
Полтавська	0.152	0.061
Рівненська	0.010	0.034
Сумська	0.059	0.279
Тернопільська	0.818	0.816
Харківська	0.043	0.069
Херсонська	0.978	0.990
Хмельницька	0.956	0.968
Черкаська	0.280	0.270
Чернівецька	0.296	0.194
Чернігівська	0.010	0.010

3.1.7 Part 1 comparison table

```r
part1_table <- tibble(
  Model = c("Baseline (no trend)", "+ Common trend (t)", "+ Region trends (t × region)", "+ Lagged loss
  N = c(nobs(m_base), nobs(m_trend), nobs(m_region_trend), nobs(m_lag)),
  Adj_R2 = c(summary(m_base)$adj.r.squared, summary(m_trend)$adj.r.squared,
             summary(m_region_trend)$adj.r.squared, summary(m_lag)$adj.r.squared),
  AIC = c(AIC(m_base), AIC(m_trend), AIC(m_region_trend), AIC(m_lag)),
  BIC = c(BIC(m_base), BIC(m_trend), BIC(m_region_trend), BIC(m_lag))
)

knitr::kable(part1_table, digits = 3, caption = "Part 1 model comparison")
```

Table 12: Part 1 model comparison

Model	N	Adj_R2	AIC	BIC
Baseline (no trend)	206	0.744	1210.085	1323.233
+ Common trend (t)	206	0.764	1193.773	1310.249
+ Region trends (t × region)	206	0.755	1217.123	1406.812
+ Lagged loss rate	182	0.733	1081.602	1193.742

Part 2 — Weather window and aggregation

3.2.0 Choosing best Part 1 specification

```
adj_r2_part1 <- c(summary(m_base)$adj.r.squared, summary(m_trend)$adj.r.squared, summary(m_region_trend
best_part1 <- c("none", "common", "region")[which.max(adj_r2_part1)]
cat("Best trend specification:", best_part1, "\n")
```

```
## Best trend specification: common
```

```
trend_controls <- switch(best_part1,
  none   = character(0),
  common = "t",
  region = c("t", region_t_cols)
)

base_controls <- c("log_sown", "war_east", dummy_cols)
all_controls  <- c(base_controls, trend_controls)
```

3.2.1 Full-year model (M_full)

```
fml_full <- as.formula(paste("yield_cpha ~", paste(c(weather_vars_full, all_controls), collapse = " + "
m_full <- lm(fml_full, data = panel_base)
summary(m_full)
```

```
##
## Call:
## lm(formula = fml_full, data = panel_base)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.8881  -1.9017   0.3164   2.2926   8.8309
##
```

```
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 15.246256  25.978913   0.587 0.558060
## days_hot_crit                0.139954   0.064748   2.162 0.032037 *
## days_cold_crit               0.118943   0.112552   1.057 0.292092
## sum_heat_pos                 0.000431   0.002342   0.184 0.854184
## sum_cold_neg                 0.002359   0.003110   0.758 0.449206
## total_precip                -0.005061   0.004785  -1.058 0.291642
## mean_wind                   -0.985369   1.061897  -0.928 0.354743
## mean_soil_moisture_pct       0.451609   0.349458   1.292 0.197983
## days_drought                -0.037829   0.020976  -1.803 0.073068 .
## log_sown                     1.858474   0.920336   2.019 0.045004 *
## war_east                    -4.508370   2.605881  -1.730 0.085411 .
## region_id                 9.374251   3.410024   2.749 0.006616 **
## region_id                 4.276318   3.588811   1.192 0.235072
## region_id            -13.397615   2.619886  -5.114 8.35e-07 ***
## region_id                 -3.906820   4.106953  -0.951 0.342802
## region_id                 6.237722   3.837247   1.626 0.105872
## region_id                -5.325921   5.181384  -1.028 0.305442
## region_id                -12.079258   4.029615  -2.998 0.003124 **
## region_id   _            9.770247   4.135254   2.363 0.019261 *
## region_id                -13.687025   2.405364  -5.690 5.36e-08 ***
## region_id                 8.362215   3.660788   2.284 0.023578 *
## region_id                 0.863174   6.651402   0.130 0.896897
## region_id                 -6.758660   4.701138  -1.438 0.152346
## region_id                 5.280707   3.579527   1.475 0.141973
## region_id                 6.342351   3.677066   1.725 0.086353 .
## region_id                 9.610969   3.510702   2.738 0.006839 **
## region_id                13.554990   3.772139   3.593 0.000426 ***
## region_id                 2.578205   3.809757   0.677 0.499482
## region_id                -17.166897   3.316537  -5.176 6.27e-07 ***
## region_id                16.733987   3.526577   4.745 4.36e-06 ***
## region_id                 8.244325   3.557889   2.317 0.021671 *
## region_id                 8.994070   4.672543   1.925 0.055895 .
## region_id                -0.166962   2.006481  -0.083 0.933780
## t                              0.604723   0.151228   3.999 9.44e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.051 on 172 degrees of freedom
## Multiple R-squared:  0.8022, Adjusted R-squared:  0.7643
## F-statistic: 21.14 on 33 and 172 DF,  p-value: < 2.2e-16
```

3.2.2 Growing season model (M_gs)

```r
weather_vars_gs <- c("days_hot_crit_gs", "days_cold_crit_gs", "sum_heat_pos_gs", "sum_cold_neg_gs",
                     "total_precip_gs", "mean_wind_gs", "mean_soil_moisture_pct", "days_drought")

fml_gs <- as.formula(paste("yield_cpha ~", paste(c(weather_vars_gs, all_controls), collapse = " + ")))
m_gs <- lm(fml_gs, data = panel_base)
summary(m_gs)
```

```
##
## Call:
## lm(formula = fml_gs, data = panel_base)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.5920  -1.8859   0.2279   2.2972   8.8724
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 62.398098  19.953493   3.127 0.002073 **
## days_hot_crit_gs             0.314153   0.090097   3.487 0.000620 ***
## days_cold_crit_gs           -0.096416   0.111773  -0.863 0.389554
## sum_heat_pos_gs             -0.008276   0.002124  -3.897 0.000139 ***
## sum_cold_neg_gs             -0.004341   0.002296  -1.890 0.060377 .
## total_precip_gs              0.014541   0.004130   3.520 0.000551 ***
## mean_wind_gs                -2.958110   0.935035  -3.164 0.001843 **
## mean_soil_moisture_pct      -0.134898   0.273788  -0.493 0.622848
## days_drought                -0.021844   0.020208  -1.081 0.281238
## log_sown                     1.777484   0.874453   2.033 0.043623 *
## war_east                    -1.337036   2.555739  -0.523 0.601543
## region_id                    3.340940   2.817114   1.186 0.237280
## region_id                   -1.967066   2.846146  -0.691 0.490414
## region_id                  -13.535530   2.083252  -6.497 8.49e-10 ***
## region_id                  -12.645240   3.147024  -4.018 8.76e-05 ***
## region_id                   -1.626276   2.978265  -0.546 0.585740
## region_id                  -15.252929   3.695800  -4.127 5.72e-05 ***
## region_id                  -16.569920   2.943593  -5.629 7.24e-08 ***
## region_id   _               2.162959   3.174493   0.681 0.496563
## region_id                  -12.122675   1.969514  -6.155 5.12e-09 ***
## region_id                    1.317019   2.844344   0.463 0.643928
## region_id                  -10.329781   4.786004  -2.158 0.032289 *
## region_id                   -8.507061   3.483863  -2.442 0.015625 *
## region_id                   -2.041822   2.817446  -0.725 0.469617
## region_id                    0.045820   2.898644   0.016 0.987406
## region_id                    1.423764   2.807200   0.507 0.612676
## region_id                    6.426874   2.964656   2.168 0.031546 *
## region_id                   -6.498989   3.062270  -2.122 0.035246 *
## region_id                  -12.283897   2.240607  -5.482 1.47e-07 ***
## region_id                   10.194721   2.838823   3.591 0.000429 ***
## region_id                    2.643254   2.783599   0.950 0.343657
## region_id                    1.621882   3.617619   0.448 0.654480
## region_id                   -0.583132   1.855186  -0.314 0.753655
## t                            0.237799   0.150437   1.581 0.115779
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.788 on 172 degrees of freedom
## Multiple R-squared:  0.827,  Adjusted R-squared:  0.7938
## F-statistic: 24.91 on 33 and 172 DF,  p-value: < 2.2e-16
```

3.2.3 Two-split model (M_split2)

```r
weather_vars_split2 <- c(
  "days_hot_crit_early", "days_cold_crit_early", "sum_heat_pos_early", "sum_cold_neg_early",
  "total_precip_early", "mean_wind_early",
  "days_hot_crit_late", "days_cold_crit_late", "sum_heat_pos_late", "sum_cold_neg_late",
  "total_precip_late", "mean_wind_late"
)

fml_split2 <- as.formula(paste("yield_cpha ~", paste(c(weather_vars_split2, all_controls), collapse = "
m_split2 <- lm(fml_split2, data = panel_base)
summary(m_split2)
```

```
##
## Call:
## lm(formula = fml_split2, data = panel_base)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.803  -1.695   0.323   2.098   8.888
##
## Coefficients: (3 not defined because of singularities)
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 55.880991  17.716244   3.154 0.001901 **
## days_hot_crit_early                NA         NA      NA       NA
## days_cold_crit_early        -0.111658   0.121763  -0.917 0.360429
## sum_heat_pos_early          -0.013897   0.003089  -4.499 1.26e-05 ***
## sum_cold_neg_early          -0.002266   0.003144  -0.721 0.472084
## total_precip_early           0.018616   0.005696   3.268 0.001308 **
## mean_wind_early             -2.730309   1.179883  -2.314 0.021854 *
## days_hot_crit_late           0.294204   0.093896   3.133 0.002034 **
## days_cold_crit_late                NA         NA      NA       NA
## sum_heat_pos_late           -0.006481   0.007884  -0.822 0.412177
## sum_cold_neg_late                  NA         NA      NA       NA
## total_precip_late            0.011888   0.005652   2.103 0.036909 *
## mean_wind_late              -0.520407   0.746216  -0.697 0.486502
## log_sown                     2.091263   0.910938   2.296 0.022906 *
## war_east                    -1.599921   2.492486  -0.642 0.521799
## region_id                    3.743665   1.953611   1.916 0.056997 .
## region_id                   -0.539278   2.032592  -0.265 0.791086
## region_id                  -13.693619   2.093395  -6.541 6.79e-10 ***
## region_id                  -12.927451   1.965494  -6.577 5.61e-10 ***
## region_id                   -0.975116   1.883683  -0.518 0.605361
## region_id                  -14.813405   2.916172  -5.080 9.82e-07 ***
## region_id                  -16.562241   2.197592  -7.537 2.69e-12 ***
## region_id   _               3.854520   2.232096   1.727 0.085999 .
## region_id                  -11.483992   2.186602  -5.252 4.43e-07 ***
## region_id                    2.866267   2.154837   1.330 0.185239
## region_id                  -11.734222   2.201571  -5.330 3.07e-07 ***
## region_id                   -9.144453   2.725474  -3.355 0.000977 ***
## region_id                   -2.001624   1.882025  -1.064 0.289033
## region_id                    1.240596   2.056487   0.603 0.547135
```

```
## region_id                 2.420631   2.040437   1.186 0.237138
## region_id              7.443274   1.944105   3.829 0.000181 ***
## region_id              -7.065266   1.998541  -3.535 0.000524 ***
## region_id             -11.751621   2.283855  -5.146 7.26e-07 ***
## region_id              11.054237   1.930038   5.727 4.49e-08 ***
## region_id                3.087313   1.878234   1.644 0.102069
## region_id               2.832787   2.361770   1.199 0.232019
## region_id             -0.251876   1.875571  -0.134 0.893329
## t                           0.314897   0.152835   2.060 0.040878 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.819 on 171 degrees of freedom
## Multiple R-squared:  0.8252, Adjusted R-squared:  0.7904
## F-statistic: 23.74 on 34 and 171 DF,  p-value: < 2.2e-16
```

3.2.4 Three-split model (M_split3)

```r
weather_vars_split3 <- c(
  "days_hot_crit_w1", "days_cold_crit_w1", "sum_heat_pos_w1", "sum_cold_neg_w1", "total_precip_w1", "mea
  "days_hot_crit_w2", "days_cold_crit_w2", "sum_heat_pos_w2", "sum_cold_neg_w2", "total_precip_w2", "mea
  "days_hot_crit_w3", "days_cold_crit_w3", "sum_heat_pos_w3", "sum_cold_neg_w3", "total_precip_w3", "mea
)

fml_split3 <- as.formula(paste("yield_cpha ~", paste(c(weather_vars_split3, all_controls), collapse = "
m_split3 <- lm(fml_split3, data = panel_base)
summary(m_split3)
```

```
##
## Call:
## lm(formula = fml_split3, data = panel_base)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.0773  -2.0191   0.1762   2.1874   8.3417
##
## Coefficients: (4 not defined because of singularities)
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 43.785929  20.117428   2.177 0.030929 *
## days_hot_crit_w1                   NA         NA      NA       NA
## days_cold_crit_w1           -0.312421   0.192753  -1.621 0.106952
## sum_heat_pos_w1             -0.012317   0.006305  -1.953 0.052457 .
## sum_cold_neg_w1             -0.007087   0.004115  -1.722 0.086867 .
## total_precip_w1              0.016086   0.008630   1.864 0.064084 .
## mean_wind_w1                -0.532287   0.765166  -0.696 0.487621
## days_hot_crit_w2                   NA         NA      NA       NA
## days_cold_crit_w2            0.335691   0.205085   1.637 0.103558
## sum_heat_pos_w2             -0.024316   0.008257  -2.945 0.003695 **
## sum_cold_neg_w2              0.017476   0.007542   2.317 0.021713 *
## total_precip_w2              0.024952   0.013341   1.870 0.063193 .
## mean_wind_w2                -0.579626   0.829041  -0.699 0.485435
```

```
## days_hot_crit_w3             0.239726   0.107747   2.225 0.027436 *
## days_cold_crit_w3                  NA         NA      NA       NA
## sum_heat_pos_w3              0.003012   0.008594   0.350 0.726453
## sum_cold_neg_w3                    NA         NA      NA       NA
## total_precip_w3              0.017216   0.006384   2.697 0.007726 **
## mean_wind_w3                -1.347026   0.799124  -1.686 0.093746 .
## log_sown                     1.708251   0.943072   1.811 0.071892 .
## war_east                    -1.125196   2.506944  -0.449 0.654138
## region_id                 4.306494   1.955814   2.202 0.029052 *
## region_id                -1.361151   2.105565  -0.646 0.518878
## region_id            -14.175067   2.252400  -6.293 2.66e-09 ***
## region_id               -13.561632   2.033923  -6.668 3.68e-10 ***
## region_id               -0.962858   1.916087  -0.503 0.615973
## region_id              -14.778151   3.109994  -4.752 4.34e-06 ***
## region_id              -17.899734   2.346319  -7.629 1.75e-12 ***
## region_id   _           3.485719   2.290818   1.522 0.130011
## region_id              -12.655291   2.303859  -5.493 1.46e-07 ***
## region_id                2.046661   2.278612   0.898 0.370377
## region_id              -12.834447   2.337247  -5.491 1.47e-07 ***
## region_id               -11.958986   2.936908  -4.072 7.20e-05 ***
## region_id               -2.251380   1.930650  -1.166 0.245236
## region_id                0.453213   2.146620   0.211 0.833046
## region_id                 2.690909   2.156417   1.248 0.213839
## region_id               7.517671   1.983799   3.790 0.000211 ***
## region_id               -6.523362   2.035151  -3.205 0.001618 **
## region_id              -13.162507   2.483777  -5.299 3.65e-07 ***
## region_id               11.538886   1.959728   5.888 2.11e-08 ***
## region_id                2.490998   1.890380   1.318 0.189412
## region_id                1.733187   2.346772   0.739 0.461229
## region_id               0.065733   1.922556   0.034 0.972766
## t                            0.259038   0.234497   1.105 0.270908
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.812 on 166 degrees of freedom
## Multiple R-squared:  0.831,  Adjusted R-squared:  0.7913
## F-statistic: 20.93 on 39 and 166 DF,  p-value: < 2.2e-16
```

3.2.5 Part 2 comparison table

```r
part2_table <- tibble(
  Model = c("M_full (full year)", "M_gs (growing season)", "M_split2 (early + late)", "M_split3 (3 wind
  N = c(nobs(m_full), nobs(m_gs), nobs(m_split2), nobs(m_split3)),
  Adj_R2 = c(summary(m_full)$adj.r.squared, summary(m_gs)$adj.r.squared,
             summary(m_split2)$adj.r.squared, summary(m_split3)$adj.r.squared),
  AIC = c(AIC(m_full), AIC(m_gs), AIC(m_split2), AIC(m_split3)),
  BIC = c(BIC(m_full), BIC(m_gs), BIC(m_split2), BIC(m_split3))
)

knitr::kable(part2_table, digits = 3, caption = "Part 2 model comparison")
```

Table 13: Part 2 model comparison

Model	N	Adj_R2	AIC	BIC
M_full (full year)	206	0.764	1193.773	1310.249
M_gs (growing season)	206	0.794	1166.204	1282.680
M_split2 (early + late)	206	0.790	1170.333	1290.136
M_split3 (3 windows)	206	0.791	1173.402	1309.845

Part 3 — Soil variables

3.3.0 Theoretical background

Why region dummies eliminate time-invariant variables

In a model with explicit region dummy variables, each dummy captures the full average level of yield for that oblast, absorbing all time-invariant characteristics — including soil properties. Formally, for any time-invariant variable z_i (such as soil organic matter), its value is identical across all T observations belonging to region i. This means z_i is a linear combination of the region dummies:

$$z_i = \sum_j z_j \cdot \mathbf{1}[\text{region} = j]$$

Including z_i alongside the region dummies therefore creates **perfect multicollinearity** — the design matrix is rank-deficient and the soil coefficient γ cannot be separately identified from the region intercepts. R will silently drop the collinear columns.

This is not a failure of the model — it is a feature. The region dummies are doing exactly what we want: controlling for all stable regional differences including soil, climate, infrastructure, and management culture. The cost is that we cannot estimate the direct effect of soil on yield.

Random effects model

The random effects (RE) model offers a way to recover soil coefficients by treating region-level unobservables as random draws from a distribution rather than fixed parameters to be estimated:

$$y_{it} = \mathbf{x}_{it}'\beta + \mathbf{z}_i'\gamma + u_i + \varepsilon_{it}$$

where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is the region-level random effect and $\varepsilon_{it} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is the idiosyncratic error. The key assumption is **orthogonality**: Cov$(u_i, \mathbf{x}_{it}) = 0$ and Cov$(u_i, \mathbf{z}_i) = 0$. Under this assumption, γ — the soil coefficients — is consistently estimated.

However, this assumption is likely violated in our setting. Soil quality is geographically correlated with climate: southern oblasts tend to have both chernozem soils (high organic matter) and hotter, drier weather. This means Cov$(u_i, \mathbf{x}_{it}) \neq 0$, and RE estimates of soil effects will absorb some of the omitted weather-climate variation, biasing γ.

The Mundlak-Chamberlain device

The Mundlak (1978) and Chamberlain (1982) approach offers a middle ground between region dummies and pure RE. It augments the RE model by including the within-region means $\bar{\mathbf{x}}_i$ of all time-varying regressors as additional controls:

$$y_{it} = \mathbf{x}_{it}'\beta + \mathbf{z}_i'\gamma + \bar{\mathbf{x}}_i'\lambda + a_i + \varepsilon_{it}$$

where a_i is now assumed orthogonal to both $\mathbf{x}_{it} - \bar{\mathbf{x}}_i$ and $\mathbf{z}_i$. Three key results follow:

1. The β estimates are numerically identical to the within (FE/dummy) estimator — time-varying weather effects are consistently estimated regardless of whether RE assumptions hold
2. The γ estimates recover soil coefficients **conditional on the long-run average weather environment** of each oblast, captured by $\bar{\mathbf{x}}_i$. This controls for the climate-soil correlation that would otherwise bias RE
3. A joint Wald test on $\lambda = \mathbf{0}$ is a Hausman-type specification test: if the Mundlak terms are jointly insignificant, RE orthogonality holds and plain RE is valid; if rejected, Mundlak is preferred

In our context, $\bar{\mathbf{x}}_i$ captures the long-run average weather of each oblast — essentially its climate. Including these means allows soil coefficients to be interpreted as the effect of soil quality **conditional on local climate**, which is the variation we want to identify.

Soil $\times$ weather interactions under region dummies

A third approach identifies soil effects through their **interaction** with time-varying weather variables. For example:

$$y_{it} = \cdots + \beta_1 \cdot \text{hot_days}_{it} + \beta_2 \cdot \text{OM}_i + \beta_3 \cdot (\text{hot_days}_{it} \times \text{OM}_i) + \text{region dummies} + \varepsilon_{it}$$

The main effect β_2 of OM_i is collinear with the region dummies and is dropped. However, the interaction term β_3 **is identified** because $\text{hot_days}_{it} \times \text{OM}_i$ varies over time (hot days change each year) even though OM_i is fixed. This coefficient answers a more policy-relevant question: **does having higher organic matter change how sensitive yield is to heat or drought stress?** If $\beta_3 > 0$ for `organic_matter × days_hot_crit`, then oblasts with richer soils suffer less yield loss per additional hot day — a direct argument for soil investment as climate adaptation.

3.3.1 Random effects model

We estimate a random effects model using `lme4::lmer` with soil variables as direct regressors alongside the best weather specification from Part 2. We use the growing season weather variables (`_gs`) as the weather block, since the Part 2 comparison will determine the best window — the growing season model is a reasonable default and avoids overfitting from the split models.

Note that soil moisture variables (`mean_soil_moisture_pct`, `days_drought`) are time-varying and are included as normal weather regressors. Only the static soil properties from `soil_data.csv` (pH, organic matter, etc.) face the collinearity problem and require the RE/Mundlak approach.

```r
oblast_map <- c(
  "Vinnytska"        = "     ",
  "Volynska"         = "     ",
  "Dnipropetrovska"  = "         ",
  "Donetska"         = "     ",
  "Zhytomyrska"      = "      ",
  "Zakarpatska"      = "       ",
  "Zaporizka"        = "      ",
  "Ivano_Frankivska" = "   _       ",
  "Kyivska"          = "     ",
  "Kirovohradska"    = "        ",
  "Luhanska"         = "     ",
  "Lvivska"          = "     ",
  "Mykolaivska"      = "       ",
  "Odeska"           = "    ",
  "Poltavska"        = "      ",
  "Rivnenska"        = "      ",
  "Sumska"           = "    ",
  "Ternopilska"      = "        ",
  "Kharkivska"       = "      ",
  "Khersonska"       = "      ",
  "Khmelnytska"      = "      ",
  "Cherkaska"        = "     ",
  "Chernivetska"     = "      ",
  "Chernihivska"     = "       "
)

soil_raw <- read_csv("soil_data.csv", show_col_types = FALSE) |>
  mutate(region = oblast_map[oblast]) |>
  dplyr::select(-oblast)

panel_base <- panel_base |>
  left_join(soil_raw, by = "region")

panel_base$region <- trimws(panel_base$region)
soil_raw$region   <- trimws(soil_raw$region)

panel_base$region <- enc2utf8(panel_base$region)
soil_raw$region   <- enc2utf8(soil_raw$region)

panel_base <- panel_base |>
  left_join(soil_raw, by = "region")

names(panel_base)
```

```
##   [1] "region"                        "year"
##   [3] "yield_cpha"                    "sown_ha"
##   [5] "harv_ha"                       "region_lat"
##   [7] "region_soil"                   "days_hot_crit"
##   [9] "days_cold_crit"                "sum_heat_pos"
##  [11] "sum_cold_neg"                  "total_precip"
##  [13] "mean_wind"                     "days_hot_crit_aug"
##  [15] "days_cold_crit_aug"            "sum_heat_pos_aug"
```

```
##  [17] "sum_cold_neg_aug"             "total_precip_aug"
##  [19] "mean_wind_aug"                "days_hot_crit_gs"
##  [21] "days_cold_crit_gs"            "sum_heat_pos_gs"
##  [23] "sum_cold_neg_gs"              "total_precip_gs"
##  [25] "mean_wind_gs"                 "days_hot_crit_early"
##  [27] "days_cold_crit_early"         "sum_heat_pos_early"
##  [29] "sum_cold_neg_early"           "total_precip_early"
##  [31] "mean_wind_early"              "days_hot_crit_late"
##  [33] "days_cold_crit_late"          "sum_heat_pos_late"
##  [35] "sum_cold_neg_late"            "total_precip_late"
##  [37] "mean_wind_late"               "days_hot_crit_w1"
##  [39] "days_cold_crit_w1"            "sum_heat_pos_w1"
##  [41] "sum_cold_neg_w1"              "total_precip_w1"
##  [43] "mean_wind_w1"                 "days_hot_crit_w2"
##  [45] "days_cold_crit_w2"            "sum_heat_pos_w2"
##  [47] "sum_cold_neg_w2"              "total_precip_w2"
##  [49] "mean_wind_w2"                 "days_hot_crit_w3"
##  [51] "days_cold_crit_w3"            "sum_heat_pos_w3"
##  [53] "sum_cold_neg_w3"              "total_precip_w3"
##  [55] "mean_wind_w3"                 "mean_soil_moisture_pct"
##  [57] "days_drought"                 "pH.x"
##  [59] "organic_matter.x"             "total_nitrogen.x"
##  [61] "cec.x"                        "clay.x"
##  [63] "sand.x"                       "silt.x"
##  [65] "ec.x"                         "esp.x"
##  [67] "c_n_ratio.x"                  "war_east"
##  [69] "log_yield"                    "log_sown"
##  [71] "t"                            "region_id"
##  [73] "loss_lag"                     "region_id     "
##  [75] "region_id     "           "region_id         "
##  [77] "region_id    "            "region_id      "
##  [79] "region_id       "        "region_id     "
##  [81] "region_id   _       "   "region_id     "
##  [83] "region_id     "           "region_id       "
##  [85] "region_id    "             "region_id      "
##  [87] "region_id     "          "region_id    "
##  [89] "region_id       "       "region_id     "
##  [91] "region_id     "          "region_id      "
##  [93] "region_id     "           "region_id      "
##  [95] "region_id       "        "region_id     _t"
##  [97] "region_id     _t"         "region_id         _t"
##  [99] "region_id    _t"          "region_id      _t"
## [101] "region_id       _t"      "region_id     _t"
## [103] "region_id   _       _t" "region_id     _t"
## [105] "region_id     _t"         "region_id       _t"
## [107] "region_id    _t"           "region_id      _t"
## [109] "region_id     _t"        "region_id    _t"
## [111] "region_id       _t"     "region_id     _t"
## [113] "region_id     _t"        "region_id      _t"
## [115] "region_id     _t"         "region_id      _t"
## [117] "region_id       _t"      "resid_base"
## [119] "resid_trend"                 "pH.y"
## [121] "organic_matter.y"            "total_nitrogen.y"
## [123] "cec.y"                       "clay.y"
```

```
## [125] "sand.y"                        "silt.y"
## [127] "ec.y"                          "esp.y"
## [129] "c_n_ratio.y"                   "pH"
## [131] "organic_matter"                "total_nitrogen"
## [133] "cec"                           "clay"
## [135] "sand"                          "silt"
## [137] "ec"                            "esp"
## [139] "c_n_ratio"
```

```
soil_vars <- c("pH", "organic_matter", "total_nitrogen", "cec", "clay", "sand")

re_weather <- c("days_hot_crit_gs", "days_cold_crit_gs",
                "sum_heat_pos_gs", "sum_cold_neg_gs",
                "total_precip_gs", "mean_wind_gs",
                "mean_soil_moisture_pct", "days_drought")

re_formula <- as.formula(paste(
  "yield_cpha ~",
  paste(c(re_weather, soil_vars, "log_sown", "war_east"), collapse = " + "),
  "+ (1 | region) + (1 | year)"
))

m_re <- lme4::lmer(re_formula, data = panel_base, REML = FALSE)
summary(m_re)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: yield_cpha ~ days_hot_crit_gs + days_cold_crit_gs + sum_heat_pos_gs +
##     sum_cold_neg_gs + total_precip_gs + mean_wind_gs + mean_soil_moisture_pct +
##     days_drought + pH + organic_matter + total_nitrogen + cec +
##     clay + sand + log_sown + war_east + (1 | region) + (1 | year)
##    Data: panel_base
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    1221.6    1288.2    -590.8    1181.6       186
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2414 -0.4704  0.0491  0.6097  2.6246
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  region   (Intercept) 14.166   3.764
##  year     (Intercept)  1.929   1.389
##  Residual             13.139   3.625
## Number of obs: 206, groups:  region, 23; year, 9
##
## Fixed effects:
##                          Estimate Std. Error t value
## (Intercept)            398.010028  70.062688   5.681
## days_hot_crit_gs         0.207845   0.093428   2.225
## days_cold_crit_gs       -0.165851   0.119215  -1.391
## sum_heat_pos_gs         -0.008871   0.002802  -3.166
## sum_cold_neg_gs         -0.006338   0.003482  -1.820
## total_precip_gs          0.013496   0.003971   3.398
```

```
## mean_wind_gs            -1.232547   0.949978  -1.297
## mean_soil_moisture_pct  -0.467172   0.213473  -2.188
## days_drought            -0.036322   0.018607  -1.952
## pH                     -19.452777   6.271630  -3.102
## organic_matter          -1.617398  17.083054  -0.095
## total_nitrogen         -19.745111  31.584211  -0.625
## cec                      1.267110   1.611605   0.786
## clay                    -3.355847   1.179737  -2.845
## sand                    -4.105924   1.013080  -4.053
## log_sown                 2.413920   0.808745   2.985
## war_east                -0.130355   2.491658  -0.052
```

```r
soil_coefs_re <- coef(summary(m_re)) |>
  as.data.frame() |>
  rownames_to_column("variable") |>
  filter(variable %in% soil_vars) |>
  rename(std_error = `Std. Error`, t_value = `t value`) |>
  mutate(
    p_approx    = 2 * pnorm(abs(t_value), lower.tail = FALSE),
    significant = case_when(
      p_approx < 0.01 ~ "***",
      p_approx < 0.05 ~ "**",
      p_approx < 0.10 ~ "*",
      TRUE            ~ "ns"
    ),
    Estimate  = round(Estimate,  3),
    std_error = round(std_error, 3)
  ) |>
  dplyr::select(variable, Estimate, std_error, t_value, p_approx, significant)

knitr::kable(soil_coefs_re, digits = 3,
  caption = "RE model - soil variable coefficients (approximate p-values via normal distribution)")
```

Table 14: RE model — soil variable coefficients (approximate p-values via normal distribution)

variable	Estimate	std_error	t_value	p_approx	significant
pH	-19.453	6.272	-3.102	0.002	***
organic_matter	-1.617	17.083	-0.095	0.925	ns
total_nitrogen	-19.745	31.584	-0.625	0.532	ns
cec	1.267	1.612	0.786	0.432	ns
clay	-3.356	1.180	-2.845	0.004	***
sand	-4.106	1.013	-4.053	0.000	***

```r
cat("\nJoint Wald test for all soil variables (RE model):\n")
```

```
##
## Joint Wald test for all soil variables (RE model):
```

```r
soil_joint_re <- multcomp::glht(m_re, linfct = paste0(soil_vars, " = 0"))
summary(soil_joint_re)
```

```
##
##   Simultaneous Tests for General Linear Hypotheses
##
## Fit: lme4::lmer(formula = yield_cpha ~ days_hot_crit_gs + days_cold_crit_gs +
##     sum_heat_pos_gs + sum_cold_neg_gs + total_precip_gs + mean_wind_gs +
##     mean_soil_moisture_pct + days_drought + pH + organic_matter +
##     total_nitrogen + cec + clay + sand + log_sown + war_east +
##     (1 | region) + (1 | year), data = panel_base, REML = FALSE)
##
## Linear Hypotheses:
##                      Estimate Std. Error z value Pr(>|z|)
## pH == 0               -19.453      6.272  -3.102   0.0103 *
## organic_matter == 0    -1.617     17.083  -0.095   1.0000
## total_nitrogen == 0   -19.745     31.584  -0.625   0.9709
## cec == 0                1.267      1.612   0.786   0.9273
## clay == 0              -3.356      1.180  -2.845   0.0237 *
## sand == 0              -4.106      1.013  -4.053   <0.001 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## (Adjusted p values reported -- single-step method)
```

3.3.2 Mundlak-Chamberlain device

We compute within-region means of all time-varying weather regressors and add them to the RE model. The Wald test on these mean terms serves as a Hausman-type specification test.

```r
mundlak_vars <- re_weather

mundlak_means <- panel_base |>
  group_by(region) |>
  summarise(
    across(all_of(mundlak_vars),
           ~ mean(.x, na.rm = TRUE),
           .names = "{.col}_mean"),
    .groups = "drop"
  )

panel_mundlak <- panel_base |>
  left_join(mundlak_means, by = "region")

mundlak_mean_cols <- paste0(mundlak_vars, "_mean")

mundlak_formula <- as.formula(paste(
  "yield_cpha ~",
  paste(c(re_weather, soil_vars, mundlak_mean_cols,
          "log_sown", "war_east"), collapse = " + "),
```

```r
  "+ (1 | region) + (1 | year)"
))

m_mundlak <- lmer(mundlak_formula, data = panel_mundlak, REML = FALSE)
summary(m_mundlak)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: yield_cpha ~ days_hot_crit_gs + days_cold_crit_gs + sum_heat_pos_gs +
##     sum_cold_neg_gs + total_precip_gs + mean_wind_gs + mean_soil_moisture_pct +
##     days_drought + pH + organic_matter + total_nitrogen + cec +
##     clay + sand + days_hot_crit_gs_mean + days_cold_crit_gs_mean +
##     sum_heat_pos_gs_mean + sum_cold_neg_gs_mean + total_precip_gs_mean +
##     mean_wind_gs_mean + mean_soil_moisture_pct_mean + days_drought_mean +
##     log_sown + war_east + (1 | region) + (1 | year)
##    Data: panel_mundlak
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    1189.2    1282.4    -566.6    1133.2       178
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.99579 -0.54833  0.08204  0.59414  2.18890
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  region   (Intercept)  1.0669  1.0329
##  year     (Intercept)  0.5495  0.7413
##  Residual             13.1182  3.6219
## Number of obs: 206, groups:  region, 23; year, 9
##
## Fixed effects:
##                              Estimate Std. Error t value
## (Intercept)                310.227789  62.721510   4.946
## days_hot_crit_gs             0.314292   0.089075   3.528
## days_cold_crit_gs           -0.173253   0.105374  -1.644
## sum_heat_pos_gs             -0.007449   0.002391  -3.115
## sum_cold_neg_gs             -0.005042   0.002721  -1.853
## total_precip_gs              0.015431   0.003825   4.034
## mean_wind_gs                -2.794304   1.007216  -2.774
## mean_soil_moisture_pct      -0.126521   0.278925  -0.454
## days_drought                -0.017943   0.019622  -0.914
## pH                         -12.097789   6.441508  -1.878
## organic_matter             -25.341796  11.113088  -2.280
## total_nitrogen              47.011414  25.453345   1.847
## cec                          0.778527   0.745344   1.045
## clay                        -0.866552   0.792217  -1.094
## sand                        -2.033019   0.496888  -4.092
## days_hot_crit_gs_mean       -0.561039   0.511127  -1.098
## days_cold_crit_gs_mean       0.582624   0.724413   0.804
## sum_heat_pos_gs_mean        -0.039818   0.021590  -1.844
## sum_cold_neg_gs_mean         0.046634   0.019362   2.408
## total_precip_gs_mean        -0.029631   0.011651  -2.543
## mean_wind_gs_mean            2.883833   1.563413   1.845
```

```
## mean_soil_moisture_pct_mean   0.157949   0.551865   0.286
## days_drought_mean             0.015304   0.056872   0.269
## log_sown                      1.557981   0.779375   1.999
## war_east                     -0.979210   2.452600  -0.399
```

```r
soil_coefs_mundlak <- coef(summary(m_mundlak)) |>
  as.data.frame() |>
  rownames_to_column("variable") |>
  filter(variable %in% soil_vars) |>
  rename(std_error = `Std. Error`, t_value = `t value`) |>
  mutate(
    p_approx    = 2 * pnorm(abs(t_value), lower.tail = FALSE),
    significant = case_when(
      p_approx < 0.01 ~ "***",
      p_approx < 0.05 ~ "**",
      p_approx < 0.10 ~ "*",
      TRUE            ~ "ns"
    ),
    Estimate  = round(Estimate,  3),
    std_error = round(std_error, 3)
  ) |>
  dplyr::select(variable, Estimate, std_error, t_value, p_approx, significant)

knitr::kable(soil_coefs_mundlak, digits = 3,
  caption = "Mundlak-Chamberlain model - soil variable coefficients")
```

Table 15: Mundlak-Chamberlain model — soil variable coefficients

variable	Estimate	std_error	t_value	p_approx	significant
pH	-12.098	6.442	-1.878	0.060	*
organic_matter	-25.342	11.113	-2.280	0.023	**
total_nitrogen	47.011	25.453	1.847	0.065	*
cec	0.779	0.745	1.045	0.296	ns
clay	-0.867	0.792	-1.094	0.274	ns
sand	-2.033	0.497	-4.092	0.000	***

```r
cat("\nHausman-type test: joint significance of Mundlak mean terms\n")
```

```
##
## Hausman-type test: joint significance of Mundlak mean terms
```

```r
cat("H :   = 0 (RE orthogonality holds, plain RE is consistent)\n")
```

```
## H :   = 0 (RE orthogonality holds, plain RE is consistent)
```

```r
cat("H :    0 (RE is biased, Mundlak correction is needed)\n\n")
```

```
## H :    0 (RE is biased, Mundlak correction is needed)
```

```r
mundlak_joint <- glht(m_mundlak, linfct = paste0(mundlak_mean_cols, " = 0"))
summary(mundlak_joint)
```

```
##
##   Simultaneous Tests for General Linear Hypotheses
##
## Fit: lmer(formula = yield_cpha ~ days_hot_crit_gs + days_cold_crit_gs +
##     sum_heat_pos_gs + sum_cold_neg_gs + total_precip_gs + mean_wind_gs +
##     mean_soil_moisture_pct + days_drought + pH + organic_matter +
##     total_nitrogen + cec + clay + sand + days_hot_crit_gs_mean +
##     days_cold_crit_gs_mean + sum_heat_pos_gs_mean + sum_cold_neg_gs_mean +
##     total_precip_gs_mean + mean_wind_gs_mean + mean_soil_moisture_pct_mean +
##     days_drought_mean + log_sown + war_east + (1 | region) +
##     (1 | year), data = panel_mundlak, REML = FALSE)
##
## Linear Hypotheses:
##                                  Estimate Std. Error z value Pr(>|z|)
## days_hot_crit_gs_mean == 0       -0.56104    0.51113  -1.098    0.814
## days_cold_crit_gs_mean == 0       0.58262    0.72441   0.804    0.946
## sum_heat_pos_gs_mean == 0        -0.03982    0.02159  -1.844    0.313
## sum_cold_neg_gs_mean == 0         0.04663    0.01936   2.408    0.093 .
## total_precip_gs_mean == 0        -0.02963    0.01165  -2.543    0.065 .
## mean_wind_gs_mean == 0            2.88383    1.56341   1.845    0.312
## mean_soil_moisture_pct_mean == 0  0.15795    0.55186   0.286    1.000
## days_drought_mean == 0            0.01530    0.05687   0.269    1.000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## (Adjusted p values reported -- single-step method)
```

```r
cat("\nAIC comparison:\n")
```

```
##
## AIC comparison:
```

```r
cat("RE model AIC:      ", round(AIC(m_re), 2), "\n")
```

```
## RE model AIC:       1221.64
```

```r
cat("Mundlak model AIC: ", round(AIC(m_mundlak), 2), "\n")
```

```
## Mundlak model AIC:  1189.25
```

```r
cat("Lower AIC is preferred.\n")
```

```
## Lower AIC is preferred.
```

3.3.3 Soil × weather interactions (OLS with region dummies)

We return to OLS with region dummies (the same framework as Parts 1 and 2) and add soil × weather interaction terms. These are identified under region dummies because the interaction varies over time. We test four theoretically motivated interactions:

- `organic_matter × days_hot_crit_gs` — does richer soil buffer heat stress?
- `organic_matter × days_drought` — does organic matter retain more plant-available water during drought?
- `clay × days_hot_crit_gs` — does clay content (higher water retention) reduce heat damage?
- `clay × days_drought` — does clay protect against drought stress?

```r
panel_interact <- panel_base |>
  mutate(
    om_x_hot       = organic_matter * days_hot_crit_gs,
    om_x_drought   = organic_matter * days_drought,
    clay_x_hot     = clay * days_hot_crit_gs,
    clay_x_drought = clay * days_drought
  )

interaction_cols <- c("om_x_hot", "om_x_drought", "clay_x_hot", "clay_x_drought")

fml_interact <- as.formula(paste(
  "yield_cpha ~",
  paste(c(re_weather, interaction_cols, all_controls), collapse = " + ")
))

m_interact <- lm(fml_interact, data = panel_interact)
summary(m_interact)
```

```
##
## Call:
## lm(formula = fml_interact, data = panel_interact)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.5186  -1.7749   0.3161   2.2844   8.8887
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 65.270236  20.482174   3.187 0.001716 **
## days_hot_crit_gs             0.070662   0.407296   0.173 0.862475
## days_cold_crit_gs           -0.094260   0.113090  -0.833 0.405746
## sum_heat_pos_gs             -0.008912   0.002169  -4.109 6.21e-05 ***
## sum_cold_neg_gs             -0.004477   0.002308  -1.939 0.054121 .
## total_precip_gs              0.015405   0.004338   3.551 0.000497 ***
## mean_wind_gs                -3.033449   0.956389  -3.172 0.001801 **
## mean_soil_moisture_pct      -0.162040   0.276669  -0.586 0.558877
## days_drought                 0.035690   0.082837   0.431 0.667131
## om_x_hot                     0.174318   0.270237   0.645 0.519771
## om_x_drought                -0.051200   0.061061  -0.839 0.402938
## clay_x_hot                  -0.023621   0.053940  -0.438 0.662013
```

```
## clay_x_drought               0.007794   0.012820   0.608 0.544049
## log_sown                     1.756445   0.900706   1.950 0.052832 .
## war_east                    -1.217740   2.651801  -0.459 0.646675
## region_id                 3.299128   2.991565   1.103 0.271687
## region_id                -2.770946   3.019840  -0.918 0.360154
## region_id           -14.612658   2.598908  -5.623 7.69e-08 ***
## region_id               -13.996702   4.097530  -3.416 0.000797 ***
## region_id               -3.943496   3.542680  -1.113 0.267239
## region_id             -16.098093   3.937517  -4.088 6.72e-05 ***
## region_id              -17.150366   3.182240  -5.389 2.36e-07 ***
## region_id   _          1.210295   3.257347   0.372 0.710690
## region_id              -11.933578   2.649455  -4.504 1.24e-05 ***
## region_id                0.394487   2.970745   0.133 0.894518
## region_id             -10.320678   5.140599  -2.008 0.046280 *
## region_id                -9.392165   3.633693  -2.585 0.010595 *
## region_id               -1.912681   3.181346  -0.601 0.548505
## region_id               -1.165765   3.116898  -0.374 0.708865
## region_id                 0.102124   3.075214   0.033 0.973548
## region_id               5.132347   3.141475   1.634 0.104186
## region_id               -6.694719   3.392577  -1.973 0.050097 .
## region_id              -12.421284   2.276532  -5.456 1.72e-07 ***
## region_id                9.488645   2.906083   3.265 0.001326 **
## region_id                2.133014   2.969147   0.718 0.473513
## region_id                0.063176   3.925208   0.016 0.987178
## region_id              -0.765103   1.882345  -0.406 0.684920
## t                            0.230971   0.155896   1.482 0.140328
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.803 on 168 degrees of freedom
## Multiple R-squared:  0.8297, Adjusted R-squared:  0.7923
## F-statistic: 22.13 on 37 and 168 DF,  p-value: < 2.2e-16
```

```r
interact_coefs <- coef(summary(m_interact)) |>
  as.data.frame() |>
  rownames_to_column("variable") |>
  filter(variable %in% interaction_cols) |>
  rename(std_error = `Std. Error`, t_value = `t value`, p_value = `Pr(>|t|)`) |>
  mutate(
    label = case_when(
      variable == "om_x_hot"       ~ "Organic matter × Hot days (GS)",
      variable == "om_x_drought"   ~ "Organic matter × Drought days",
      variable == "clay_x_hot"     ~ "Clay × Hot days (GS)",
      variable == "clay_x_drought" ~ "Clay × Drought days"
    ),
    significant = case_when(
      p_value < 0.01 ~ "***",
      p_value < 0.05 ~ "**",
      p_value < 0.10 ~ "*",
      TRUE           ~ "ns"
    ),
    Estimate  = round(Estimate,  4),
    std_error = round(std_error, 4),
    p_value   = round(p_value,   4)
```

```r
  ) |>
  dplyr::select(label, Estimate, std_error, t_value, p_value, significant)

knitr::kable(interact_coefs, digits = 4,
  caption = "Soil × weather interaction coefficients (OLS with region dummies)")
```

Table 16: Soil × weather interaction coefficients (OLS with region dummies)

label	Estimate	std_error	t_value	p_value	significant
Organic matter × Hot days (GS)	0.1743	0.2702	0.6451	0.5198	ns
Organic matter × Drought days	-0.0512	0.0611	-0.8385	0.4029	ns
Clay × Hot days (GS)	-0.0236	0.0539	-0.4379	0.6620	ns
Clay × Drought days	0.0078	0.0128	0.6079	0.5440	ns

```r
cat("\nJoint significance of all soil × weather interactions:\n")
```

```
##
## Joint significance of all soil × weather interactions:
```

```r
linearHypothesis(m_interact,
  c("om_x_hot = 0", "om_x_drought = 0", "clay_x_hot = 0", "clay_x_drought = 0"))
```

```
##
## Linear hypothesis test:
## om_x_hot = 0
## om_x_drought = 0
## clay_x_hot = 0
## clay_x_drought = 0
##
## Model 1: restricted model
## Model 2: yield_cpha ~ days_hot_crit_gs + days_cold_crit_gs + sum_heat_pos_gs +
##     sum_cold_neg_gs + total_precip_gs + mean_wind_gs + mean_soil_moisture_pct +
##     days_drought + om_x_hot + om_x_drought + clay_x_hot + clay_x_drought +
##     log_sown + war_east + region_id      + region_id      +
##     region_id          + region_id     +
##     region_id       + region_id       +
##     region_id      + region_id   _       +
##     region_id      + region_id      +
##     region_id       + region_id     +
##     region_id      + region_id      +
##     region_id     + region_id        +
##     region_id      + region_id      +
##     region_id       + region_id      +
##     region_id       + region_id       +
##     t
##
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    172 2468.5
## 2    168 2429.2  4    39.333 0.6801 0.6067
```

3.3.4 Individual and joint significance summary

```
soil_compare <- soil_coefs_re |>
  dplyr::select(variable, Estimate, p_approx, significant) |>
  rename(est_RE = Estimate, p_RE = p_approx, sig_RE = significant) |>
  left_join(
    soil_coefs_mundlak |>
      dplyr::select(variable, Estimate, p_approx, significant) |>
      rename(est_Mundlak = Estimate, p_Mundlak = p_approx, sig_Mundlak = significant),
    by = "variable"
  )

knitr::kable(soil_compare, digits = 3,
  caption = "Soil variable coefficients: RE vs Mundlak-Chamberlain comparison")
```

Table 17: Soil variable coefficients: RE vs Mundlak-Chamberlain comparison

variable	est_RE	p_RE	sig_RE	est_Mundlak	p_Mundlak	sig_Mundlak
pH	-19.453	0.002	***	-12.098	0.060	*
organic_matter	-1.617	0.925	ns	-25.342	0.023	**
total_nitrogen	-19.745	0.532	ns	47.011	0.065	*
cec	1.267	0.432	ns	0.779	0.296	ns
clay	-3.356	0.004	***	-0.867	0.274	ns
sand	-4.106	0.000	***	-2.033	0.000	***

```
cat("\n=== Joint significance summary ===\n")
```

```
##
## === Joint significance summary ===
```

```
cat("RE model - joint test p-value: see glht output above\n")
```

```
## RE model - joint test p-value: see glht output above
```

```
cat("Mundlak model - joint test p-value: see glht output above\n")
```

```
## Mundlak model - joint test p-value: see glht output above
```

```
cat("Interaction model - joint test p-value: see linearHypothesis output above\n")
```

```
## Interaction model - joint test p-value: see linearHypothesis output above
```

3.3.5 Part 3 comparison table

```r
part3_table <- tibble(
  Model = c(
    "RE (soil direct effects)",
    "Mundlak-Chamberlain (soil + climate controls)",
    "OLS + region dummies + soil × weather interactions"
  ),
  N      = c(nobs(m_re), nobs(m_mundlak), nobs(m_interact)),
  Adj_R2 = c(NA_real_, NA_real_, summary(m_interact)$adj.r.squared),
  AIC    = c(AIC(m_re), AIC(m_mundlak), AIC(m_interact)),
  BIC    = c(BIC(m_re), BIC(m_mundlak), BIC(m_interact))
)

knitr::kable(part3_table, digits = 3,
  caption = "Part 3 model comparison: three approaches to soil variable identification")
```

Table 18: Part 3 model comparison: three approaches to soil variable identification

Model	N	Adj_R2	AIC	BIC
RE (soil direct effects)	206	NA	1221.644	1288.201
Mundlak-Chamberlain (soil + climate controls)	206	NA	1189.250	1282.430
OLS + region dummies + soil × weather interactions	206	0.792	1170.895	1300.683

Note: Adj R^2 is not directly comparable between lme4 mixed models (RE, Mundlak) and OLS (interactions model). AIC and BIC are valid for comparison within each estimator class but should be interpreted with caution across classes.

3.3.6 Conclusions from soil analysis

On the choice of estimator:

The Hausman-type test on the Mundlak mean terms reveals whether the plain RE model is adequate. If the joint test on λ rejects the null, the RE orthogonality assumption fails — meaning that the long-run climate of each oblast is correlated with its unobserved random effect. In this case, the Mundlak model is preferred: it corrects for this bias by explicitly controlling for regional climate averages, and its soil coefficients should be interpreted as the effect of soil quality *conditional on the oblast's climate*.

On direct soil effects (RE and Mundlak):

The RE model provides a first approximation of which soil properties matter. Positive coefficients on `organic_matter` would confirm that higher soil carbon improves average yield — consistent with its role in water retention, nutrient supply, and microbial activity. The `clay` coefficient captures a trade-off: clay soils retain more moisture (positive in dry years) but also drain poorly and warm more slowly in spring (potentially negative overall). If the Mundlak correction substantially changes the magnitude or significance of soil coefficients relative to plain RE, this confirms that ignoring the climate-soil correlation would have led to misleading inferences.

On soil × weather interactions (OLS model):

The interaction coefficients answer the most actionable question for agribusiness: does soil quality change the *sensitivity* of yield to weather shocks? A positive and significant coefficient on `organic_matter × days_hot_crit_gs` means that each additional hot day causes less yield loss in oblasts with richer soils — direct empirical evidence for the buffering role of soil organic matter under heat stress. Similarly, a significant `organic_matter × days_drought` interaction would support the agronomic claim that organic matter improves drought resilience by enhancing water-holding capacity. If the joint test on all four interaction terms is significant, we can conclude that soil quality modulates weather risk — an important finding for climate adaptation strategy.

Limitation: With only 24 oblasts and static soil data, identification of soil effects relies entirely on cross-sectional variation. The estimates are sensitive to unobserved regional factors that co-vary with both soil quality and yield (e.g., management culture, land consolidation, irrigation access). The Mundlak correction partially addresses this but cannot eliminate all confounding. These results should be interpreted as indicative associations rather than causal estimates.

Conclusions

What the data showed

The exploratory analysis established several facts that directly shaped the modelling strategy. First, yield levels and trends differ substantially across oblasts — western regions consistently outperform eastern ones, and the patterns diverge enough that a single common trend cannot summarise all regions. This motivated including both a common time trend and, where warranted, region-specific trends. Second, the VIF analysis confirmed that solar radiation, ET_0, and dewpoint are largely redundant given the information already captured by temperature stress metrics and precipitation: removing these three variables reduced the maximum VIF from 70.25 to 6.91 with no loss of substantive content.

What the models showed

Part 1 — Stationarity and trend. Raw yield series are non-stationary in most oblasts, but model residuals show no persistent unit-root behaviour, confirming that the regression specification adequately absorbs trends. Adding a common time trend improved adjusted R^2 from 0.744 to 0.764 and reduced AIC by 16 points. Region-specific trends added no further improvement and consumed too many degrees of freedom. The lagged loss rate was individually insignificant, suggesting that farmer behavioural responses to prior-year losses are not a primary driver of subsequent yield in this dataset.

Part 2 — Weather window. The growing season model (M_gs) outperformed the full-year model on all criteria: adjusted R^2 of 0.794 versus 0.764, AIC lower by 28 points. The two-split and three-split models performed similarly to M_gs but added complexity without meaningful gain. Within the growing season specification, the key significant weather variables were: days with extreme heat stress (positive, counterintuitive — likely a proxy for summer radiation), accumulated non-critical heat (negative — sustained warmth above the growth optimum reduces yield), total growing season precipitation (positive), and mean wind speed (negative — lodging risk). These results confirm Hypothesis 3: the growing season window dominates, supporting Hypothesis 2, and extreme-day counts carry predictive content beyond mean temperature, supporting Hypothesis 3.

Part 3 — Soil variables. Three complementary approaches were used to identify soil effects that are collinear with region dummies in the OLS framework. The plain RE model identified pH, clay, and sand as individually significant. After applying the Mundlak-Chamberlain correction — which controls for the correlation between long-run regional climate and soil quality — organic matter and sand emerged as the most robust soil effects, while clay lost significance. The Hausman-type test on the Mundlak mean terms was only marginally significant for two terms (accumulated cold and precipitation), suggesting the RE orthogonality assumption is not severely violated but the Mundlak model is still preferred on AIC grounds (1189 vs 1222). The soil × weather interaction model found no statistically significant interactions between organic matter or clay and heat or drought stress (joint F-test $p = 0.61$), failing to support the buffering hypothesis with this dataset.

Answers to the research hypotheses

H1 (meteorological significance): Confirmed. Growing season temperature stress, precipitation, and wind are jointly and individually significant drivers of yield. The relationship is non-linear in the sense that both heat stress days and accumulated heat above the growth optimum matter, with opposing signs.

H2 (seasonal relevancy): Confirmed. The growing season model outperforms the full-year model. Winter weather variables, once separated into their own window, add little incremental explanatory power beyond the growing season.

H3 (extreme day classification): Confirmed. Days with extreme heat stress and accumulated non-critical heat are both significant, while raw mean temperature metrics have higher VIF and lower t-statistics. Daily data enabling extreme-event counting adds value over simple averages.

H4 (redundancy): Confirmed. Solar radiation, ET_0, and dewpoint can be dropped from the data collection protocol without loss. The mandatory set is: daily maximum and minimum temperature, daily precipitation, daily wind speed, and soil moisture.

H5 (soil as moderator): Partially confirmed. Soil organic matter and sand content have direct effects on average yield (identified through the Mundlak model), but do not significantly moderate weather sensitivity in the interaction model. Soil quality matters for baseline productivity but does not appear to buffer heat or drought shocks in this 9-year panel.

Practical recommendation for agribusinesses

The minimum sufficient data collection protocol for yield forecasting at oblast level consists of: daily Tmax and Tmin (to construct extreme-day counts and accumulated heat/cold), daily precipitation totals, daily wind gusts, and periodic soil moisture readings at 0–7 cm depth. Solar radiation, ET_0, dewpoint, and cloudiness are redundant given this set and need not be collected. Soil properties from a single baseline survey provide useful information about regional yield potential but should not be expected to improve short-run weather-driven forecasts.